CHAPTER 12

Separations and Purifications

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 33a5800915e64c27d3484261ff203e1d62548d8f62250496ec6f30fd68e4fe6d

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

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ch12:exercise:choose-distillation | choose-distillation:stage-1, choose-distillation:stage-2, choose-distillation:stage-3

ch12:exercise:rf-calculation | rf-calculation:stage-1, rf-calculation:stage-2, rf-calculation:stage-3

ch12:exercise:chromatography-match | chromatography-match:stage-1, chromatography-match:stage-2, chromatography-match:stage-3

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ch12:molecule:benzoicacid | benzoicacid

ch12:molecule:benzoate | benzoate

ch12:molecule:toluene | toluene

ch12:molecule:glycylglycine | glycylglycine

ch12:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 12.1 Solubility-Based Methods

scope-2: 12.2 Distillation

scope-3: 12.3 Chromatography

Objective: Plan acid–base extraction and crystallization

Objective: Choose distillation conditions

Objective: Calculate Rf and match chromatography methods

Supplied Kaplan curriculum: Chapter 12; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter12b.xhtml

Size exclusion principles: Waters educational primer | https://www.waters.com/nextgen/us/en/education/primers/beginners-guide-to-size-exclusion-chromatography/gpc-basic-chemistry.html

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

solubility | Extraction, filtration, and recrystallization

Change solubility to move a compound

solubility:rule | Core Rules

Liquid–liquid extraction partitions solutes between immiscible phases. Acid/base chemistry can turn a neutral acid or base into a more water-soluble ion. Recrystallization exploits high hot-solvent and low cold-solvent solubility of the desired compound. Hot gravity filtration removes insoluble material; vacuum filtration collects crystals.

Visual card: Liquid–Liquid Acid/Base Extraction, Separatory Funnel Density & Recrystallization

Chemistry diagram: Separation apparatus schematic. Top/bottom phase identities depend on density, not on the word organic.

Carboxylic Acid (pKa 5) → Extracted by $NaHCO_3$

Phenol (pKa 10) → Extracted by NaOH

Amine (pKaH 10) → Extracted by HCl

1. Wash with 5% HCl (Acid) | Protonates Amine: RNH_2 → $RNH_3^+Cl^-$ | Amine moves into AQUEOUS layer (acid/phenol/neutral stay organic)

2. Wash with 5% $NaHCO_3$ (Weak Base) | Deprotonates Carboxylic Acid ONLY: RCOOH → $RCOO^-Na^+$ | Carboxylic acid moves to AQUEOUS layer (phenol pKa 10 stays organic!)

3. Wash with 5% NaOH (Strong Base) | Deprotonates Phenol: ArOH → ArO^-Na^+ | Phenol moves to AQUEOUS layer; Neutral stays in ORGANIC layer!

Charge Controls Layer Partitioning (Ionized = Aqueous; Neutral = Organic) — — Converting a neutral organic acid or base into an ionic salt ($RCOO^-Na^+$, ArO^-Na^+, or $RNH_3^+Cl^-$) makes it water-soluble so it moves into the aqueous layer.

Selective Base Extraction: Weak Base $NaHCO_3$ (pKaH 6.4) vs. Strong Base NaOH (pKaH 15.7) — — $NaHCO_3$ is strong enough to deprotonate carboxylic acids (pKa ~5), but TOO WEAK to deprotonate phenols (pKa ~10)! Always extract with $NaHCO_3$ first, then NaOH second.

Which Layer Is on the Bottom? Check Density (d), NOT "Organic vs. Aqueous"! — — Water has d = 1.00 g/mL. Non-chlorinated solvents (Et_2O d=0.71, hexanes d=0.66, EtOAc d=0.90) float on TOP; chlorinated solvents (CH_2Cl_2 d=1.33, $CHCl_3$ d=1.49) sink to the BOTTOM!

solubility:cue | Memory Cues

For an acid: deprotonate to move toward water, then acidify to recover the neutral form.

Visual card: 3-Step Extraction & Recovery Flowchart

Chemistry diagram: Acid and conjugate-base graphs used in extraction; not resonance contributors.

Which component are you isolating or where is your compound located?

Separate Benzoic Acid (pKa 4.2) from Phenol (pKa 10.0) — Add Aq. $NaHCO_3$! Benzoic Acid → Aqueous; Phenol Stays Organic! — If you added NaOH first, BOTH benzoic acid and phenol would deprotonate into the same aqueous layer and fail to separate!

Recover the Neutral Solid Back from an Aqueous Extract — Acidify Basic Extract (Add HCl to $RCOO^-$/ArO^-) OR Basify Acidic Extract (Add NaOH to RNH_3^+) — Neutralizing the ionic salt restores the uncharged organic molecule, which precipitates (vacuum filter via Büchner funnel) or re-extracts into ether.

Purify an Impure Solid by Recrystallization — Choose a Solvent Where Compound Is Insoluble COLD, Soluble HOT! — Dissolve in minimum boiling solvent, cool slowly so pure crystals form while impurities stay dissolved in the mother liquor, then vacuum filter.

solubility:trap | Common Mistakes / MCAT Traps

The organic layer is not always on top; density determines position. Keep track of which layer contains the target. Several smaller extractions can outperform one using the same total solvent volume. A wash removes impurities while retaining the target in its phase.

Visual card: Assuming the Organic Layer Is Always on Top or Using NaOH Before $NaHCO_3$

Common Trap — — Draining the bottom layer of a dichloromethane (CH_2Cl_2) / water extraction thinking it is the aqueous layer.

Correct Reasoning — — Dichloromethane (CH_2Cl_2, d = 1.33 g/mL) and chloroform ($CHCl_3$, d = 1.49 g/mL) are denser than water and form the BOTTOM layer!

solubility:app | MCAT Applications

Acid/base extraction separates acidic, basic, and neutral drug-like compounds by controlling their charge and partitioning.

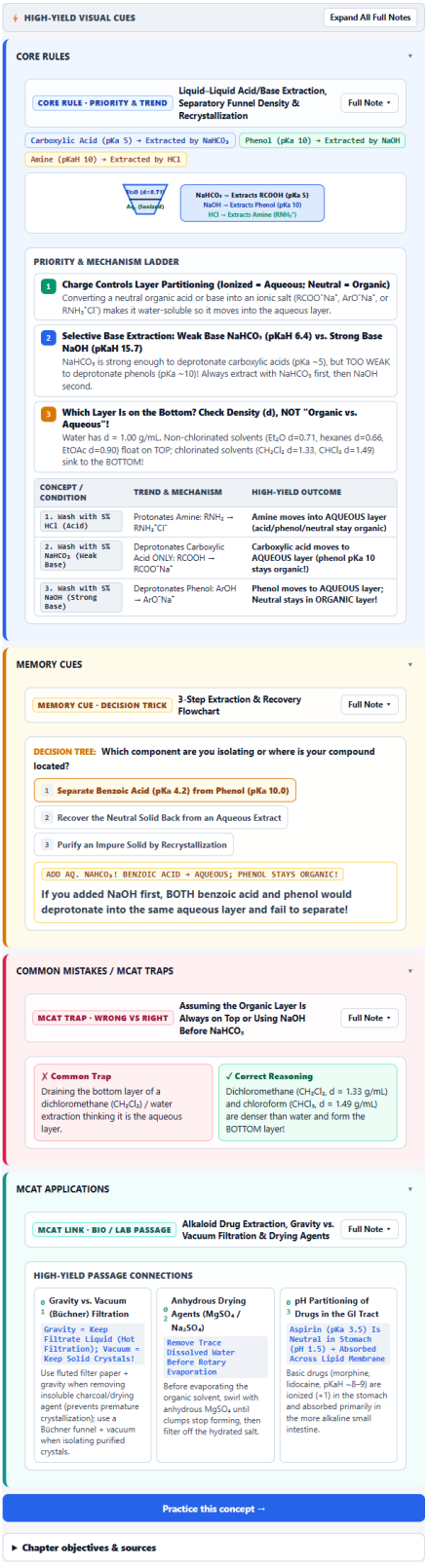
Visual card: Alkaloid Drug Extraction, Gravity vs. Vacuum Filtration & Drying Agents

Gravity vs. Vacuum (Büchner) Filtration — Gravity = Keep Filtrate Liquid (Hot Filtration); Vacuum = Keep Solid Crystals! — Use fluted filter paper + gravity when removing insoluble charcoal/drying agent (prevents premature crystallization); use a Büchner funnel + vacuum when isolating purified crystals.

Anhydrous Drying Agents ($MgSO_4$ / Na_2SO_4) — Remove Trace Dissolved Water Before Rotary Evaporation — Before evaporating the organic solvent, swirl with anhydrous $MgSO_4$ until clumps stop forming, then filter off the hydrated salt.

pH Partitioning of Drugs in the GI Tract — Aspirin (pKa 3.5) Is Neutral in Stomach (pH 1.5) → Absorbed Across Lipid Membrane — Basic drugs (morphine, lidocaine, pKaH ~8–9) are ionized (+1) in the stomach and absorbed primarily in the more alkaline small intestine.

Screenshot: solubility-visuals



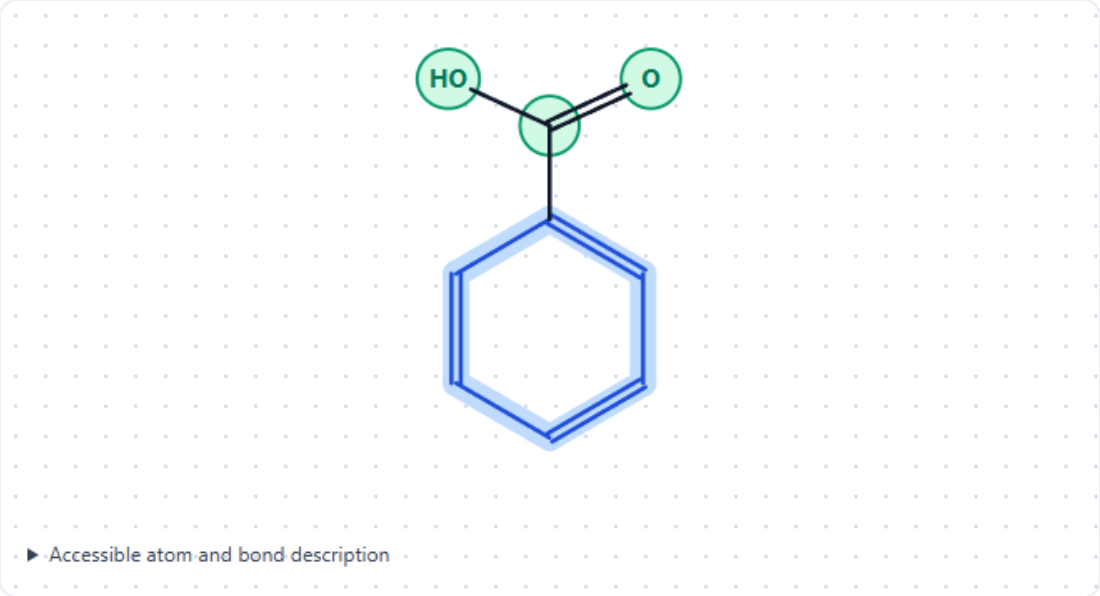
solubility:step-1 | 1. Neutral aromatic acid

Benzoic acid combines a nonpolar aromatic region with a polar acid group. Its partitioning depends on both molecular structure and ionization.

Displayed molecule ID: benzoicacid

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Screenshot: solubility-step-1



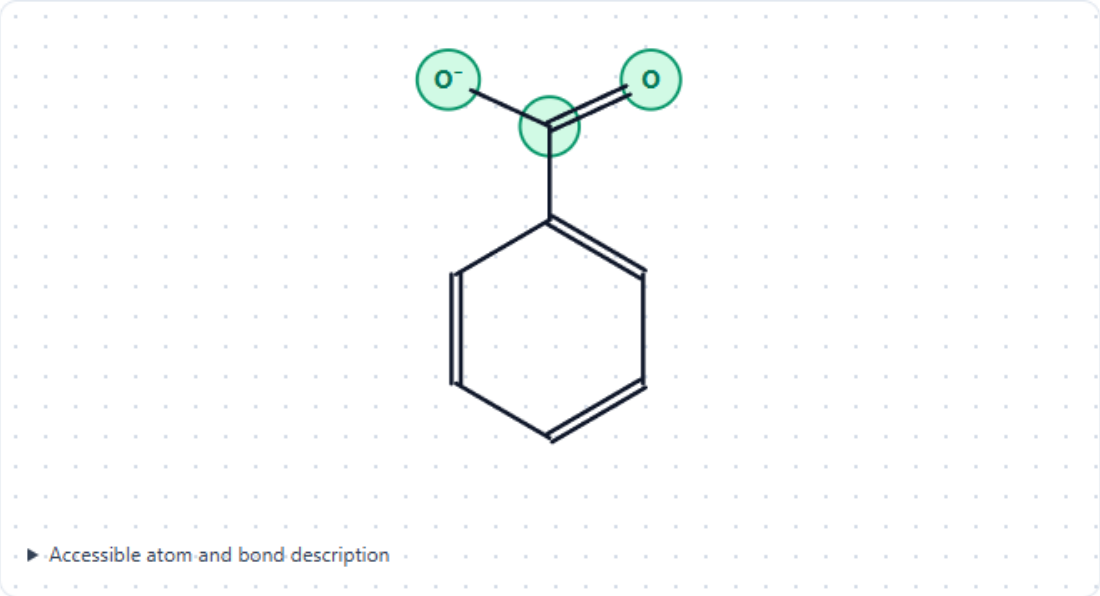
solubility:step-2 | 2. Ionize the acid

At sufficiently high pH, benzoate predominates and is generally more water soluble than neutral benzoic acid. Acidifying can regenerate neutral acid for recovery when solubility conditions permit.

Displayed molecule ID: benzoate

{"highlights": {"prin": [6, 7, 8]}, "revealName": true}

Screenshot: solubility-step-2



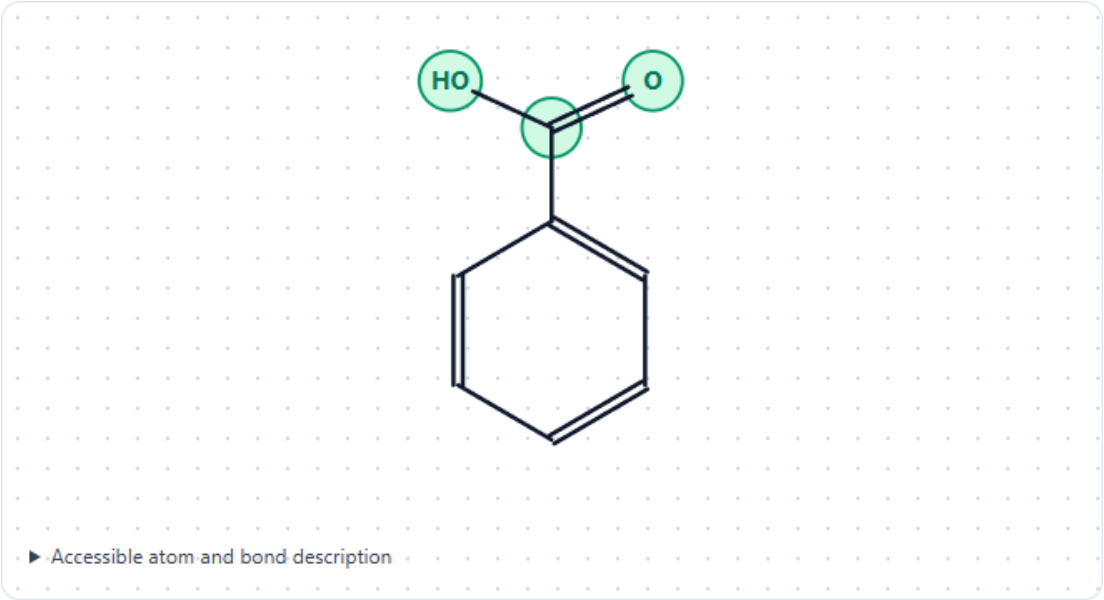
solubility:step-3 | 3. Recrystallize the recovered solid

Use a suitable solvent system in which the desired solid dissolves hot and crystallizes on cooling. Soluble impurities should remain mainly in the mother liquor; insoluble impurities can be filtered before cooling.

Displayed molecule ID: benzoicacid

{"highlights": {"prin": [6, 7, 8]}, "revealName": true}

Screenshot: solubility-step-3



distillation | Distillation and volatility

Vapor is enriched in the more volatile component

distillation:rule | Core Rules

Distillation separates liquids by volatility. Simple distillation suits sufficiently different volatilities; fractional distillation adds repeated vaporization/condensation for closer boiling mixtures. Lower pressure lowers boiling temperature, so vacuum distillation can help with high-boiling or heat-sensitive liquids. Superheating means a liquid is above its boiling temperature without sustained boiling. Nucleation sites from appropriate boiling aids or stirring can support smooth boiling; chips are not added to an already hot superheated liquid.

Visual card: Simple, Fractional, Vacuum & Steam Distillation: Raoult's Law & Theoretical Plates

Chemistry diagram: Apparatus schematic: vapor passes a thermometer at the takeoff, cools in the condenser and collects as distillate.

$P_{total} = P°_A \cdot X_A + P°_B \cdot X_B$ (Raoult's Law) | Liquid Boils When $P_{vapor} = P_{external}$

Simple: ΔBP > ~25–30 °C

Fractional: ΔBP < ~25 °C (Multiple Plates)

Vacuum: BP > 150 °C (↓ P → ↓ BP)

Steam: Immiscible Oils

Simple Distillation | ΔBP > 25–30 °C (or non-volatile impurity) | 1 vaporization–condensation cycle

Fractional Distillation | ΔBP < 25–30 °C (close boiling points) | Packed fractionating column = many theoretical plates

Vacuum Distillation | BP > 150 °C (prevents thermal decomposition) | Reduced pressure lowers atmospheric P so liquid boils at lower T

Which Compound Distills First? — — always the MORE VOLATILE component (HIGHER vapor pressure P° = LOWER boiling point = WEAKER intermolecular forces)!

Curriculum Heuristic Cutoffs (Not Rigid Physical Laws) — — Use Simple Distillation when boiling points differ widely (>25–30 °C); use Fractional Distillation when boiling points are close (<25–30 °C); use Vacuum Distillation for high-boiling (>150 °C) heat-sensitive compounds.

distillation:cue | Memory Cues

Fractionating column = repeated equilibrium stages. Vacuum = lower boiling temperature.

Visual card: Select the Right Distillation Apparatus for Any Mixture

Examine the boiling points and thermal stability of the liquids:

Close Boiling Points: Hexane (69 °C) + Ethyl Acetate (77 °C) (ΔBP = 8 °C) — Fractional Distillation (Fractionating Column Between Pot and Condenser) — Glass beads or Vigreux indentations provide large surface area for repeated vaporization-condensation cycles (theoretical plates), enriching the vapor in the lower-boiling component.

High-Boiling Compound That Decomposes at 195 °C (BP = 220 °C at 1 atm) — Vacuum (Reduced-Pressure) Distillation — Connecting a vacuum pump lowers $P_{external}$; because a liquid boils as soon as P_{vapor} equals $P_{external}$, it distills safely at a much lower temperature!

Azeotrope Mixture (e.g., 95.6% Ethanol / 4.4% Water) — Boils at a Constant Temperature Without Changing Composition! — Non-ideal H-bonding deviations from Raoult's law create a minimum- or maximum-boiling azeotrope that cannot be separated past the azeotropic ratio by ordinary distillation alone.

distillation:trap | Common Mistakes / MCAT Traps

Boiling-point cutoffs are heuristics, not universal rules. Azeotropes can limit ordinary distillation. Distillation does not guarantee a pure first fraction merely because one component has a lower boiling point. The supplied curriculum suggests simple distillation for boiling points roughly 25 °C or more apart and below about 150 °C, fractionation for closer boiling points, and considering vacuum for high normal boiling temperatures (around or above 150 °C) or heat-sensitive materials. These are selection heuristics, not guarantees; pressure, relative volatility, composition, apparatus and decomposition behavior matter.

Visual card: Saying Higher Boiling Point Means Higher Vapor Pressure or Distilling Past an Azeotrope

Common Trap — — Saying the compound with the higher boiling point has a higher vapor pressure and distills first.

Correct Reasoning — — Boiling point and vapor pressure are INVERSELY related: stronger IMFs = LOWER vapor pressure = HIGHER boiling point (stays behind in the distilling pot while the lower-BP liquid distills first)!

distillation:app | MCAT Applications

Choose a method based on volatility, thermal stability, pressure, and mixture behavior; record boiling temperatures with pressure.

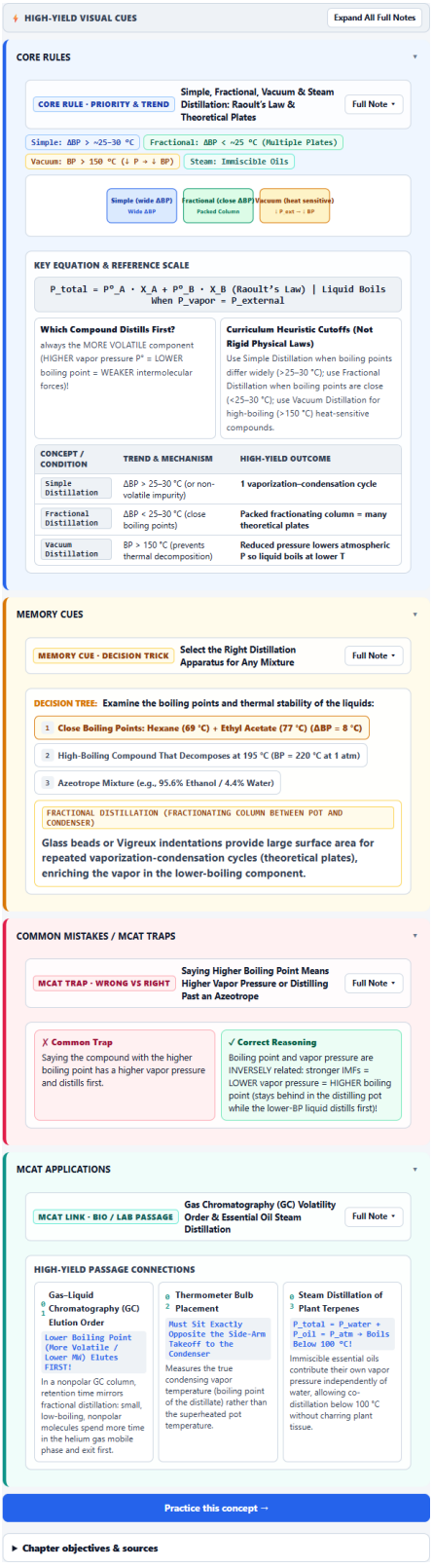
Visual card: Gas Chromatography (GC) Volatility Order & Essential Oil Steam Distillation

Gas–Liquid Chromatography (GC) Elution Order — Lower Boiling Point (More Volatile / Lower MW) Elutes FIRST! — In a nonpolar GC column, retention time mirrors fractional distillation: small, low-boiling, nonpolar molecules spend more time in the helium gas mobile phase and exit first.

Thermometer Bulb Placement — Must Sit Exactly Opposite the Side-Arm Takeoff to the Condenser — Measures the true condensing vapor temperature (boiling point of the distillate) rather than the superheated pot temperature.

Steam Distillation of Plant Terpenes — $P_{total} = P_{water} + P_{oil} = P_{atm}$ → Boils Below 100 °C! — Immiscible essential oils contribute their own vapor pressure independently of water, allowing co-distillation below 100 °C without charring plant tissue.

Screenshot: distillation-visuals



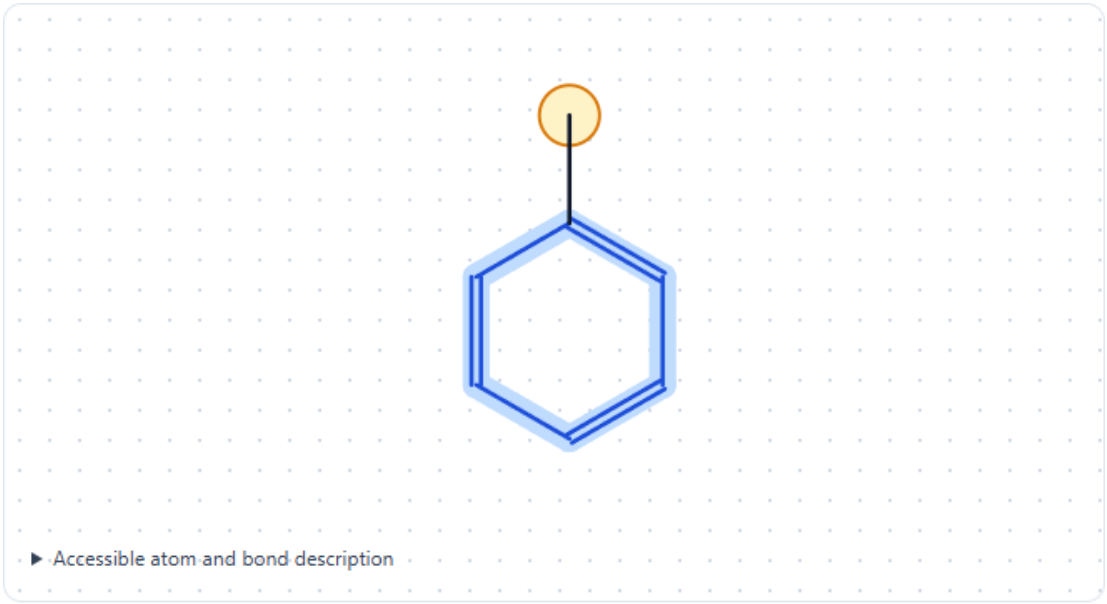
distillation:step-1 | 1. A molecular property

Toluene is a neutral volatile organic liquid. Intermolecular interactions influence its vapor pressure and boiling behavior; the structure alone does not supply an exact boiling point.

Displayed molecule ID: toluene

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Screenshot: distillation-step-1



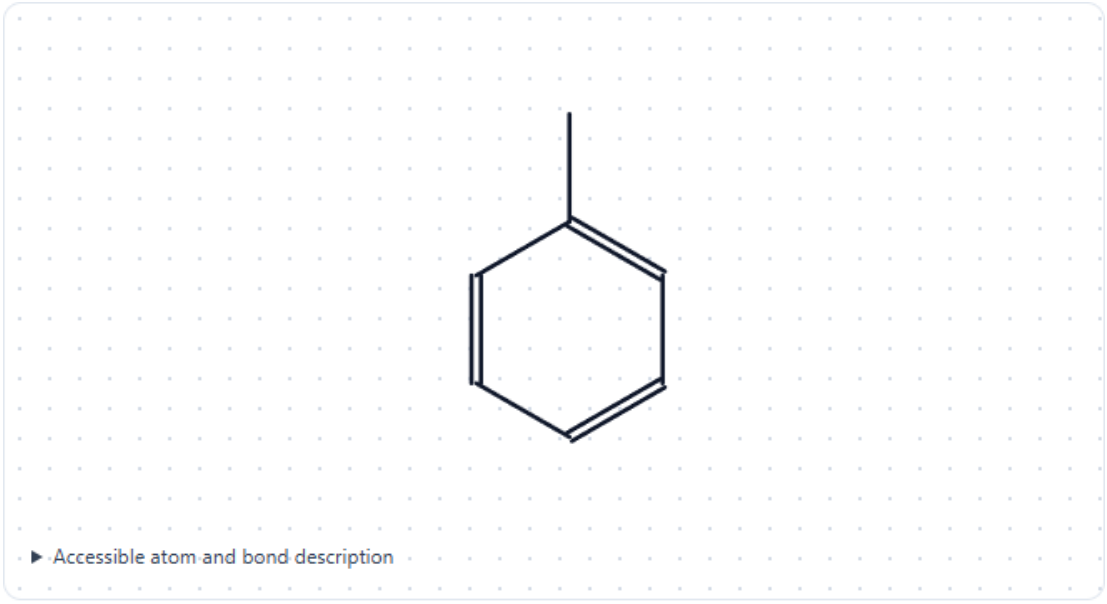
distillation:step-2 | 2. Choose a method

For a hypothetical close-boiling mixture, fractionation can improve enrichment. If the material decomposes near its atmospheric boiling point, reducing pressure may allow a lower operating temperature.

Displayed molecule ID: toluene

{"highlights": {}, "revealName": false}

Screenshot: distillation-step-2



tlc | TLC and normal-phase columns

Retention competes with the mobile phase

tlc:rule | Core Rules

In normal-phase TLC, polar silica is stationary and solvent moves upward by capillary action. Stronger attraction to silica generally lowers Rf. Rf = distance traveled by spot center / distance traveled by solvent front, both measured from the baseline. A stronger eluent usually raises Rf on silica. Paper chromatography uses polar cellulose and often an associated water layer; partitioning and adsorption depend on the solvent system. Invisible TLC spots can be visualized by UV or suitable iodine, vanillin or phosphomolybdic-acid stains; detection depends on the analyte.

Visual card: Thin-Layer Chromatography (TLC) & Column Chromatography: Normal Phase vs. Reverse Phase

Chemistry diagram: Illustrative TLC retention model with distinct spot labels; not experimental measurements or a purity guarantee.

Rf = (Distance Traveled by Spot from Baseline) / (Distance Traveled by Solvent Front from Baseline) (Always $0 \leq Rf \leq 1$)

Normal Phase: Polar Silica (SiO_2) + Nonpolar Eluent

Rf = d_spot / d_solvent

↑ Polarity → ↓ Rf (Elutes LAST)

Reverse Phase = Opposite!

Normal-Phase TLC / Column | Stationary = Polar Silica (SiO_2) / Alumina | Nonpolar moves FASTEST (↑ Rf, elutes 1st); Polar sticks (↓ Rf, elutes last)

Reverse-Phase HPLC (C18) | Stationary = Nonpolar C_{18} alkyl chains | Polar moves FASTEST (elutes 1st in H_2O/MeCN); Nonpolar sticks (elutes last)

More Polar Mobile Phase | Adding EtOAc/MeOH to hexanes on silica | Displaces compounds from silica → usually increases Rf on normal-phase silica

Normal-Phase Rule ("Like Binds Like" on Polar Silica Gel SiO_2) — — Polar compounds (carboxylic acids, diols, alcohols) H-bond tightly to polar Si–OH groups → LOW Rf & SLOW column elution. Nonpolar compounds (alkanes, alkenes, ethers) barely stick → HIGH Rf & FIRST out of the column!

Polarity Ladder (Lowest Rf → Highest Rf on Silica) — — Carboxylic Acid / Amino Acid < Diol < Alcohol < Amine < Amide < Ketone/Aldehyde < Ester < Ether < Alkene < Alkane.

tlc:cue | Memory Cues

Stronger stationary-phase binding → slower travel. Rf has no units.

Visual card: TLC Rf & Column Elution Order Predictor

What chromatographic condition is specified in the passage?

Normal-Phase Silica Gel TLC / Column (Standard Unless Stated Otherwise) — Nonpolar = HIGH Rf & Elutes FIRST; Polar = LOW Rf & Elutes LAST — Think: "Polar sticks to Polar Silica!" A spot with Rf = 0.82 is much LESS polar than a spot with Rf = 0.24.

Reverse-Phase Chromatography (C18 / RP-HPLC) — EXACT OPPOSITE! Polar Elutes FIRST; Nonpolar Sticks to C18 Tails & Elutes LAST! — In Reverse Phase, the stationary phase is greasy C_{18} hydrocarbon chains and the mobile phase is polar water/acetonitrile.

Switching Mobile Phase from 10% EtOAc/Hexanes to 50% EtOAc/Hexanes on Silica — Usually increases Rf on normal-phase silica; selectivity can change — Polar ethyl acetate competes for silica binding sites, sweeping all analytes faster with the solvent front.

tlc:trap | Common Mistakes / MCAT Traps

Rf depends on plate, solvent, and conditions; it is not a permanent molecular constant. Equal Rf is supporting evidence, not definitive identification. In reversed-phase systems the stationary phase is nonpolar, so polarity trends differ.

Visual card: Confusing High Rf with High Retention Time or Measuring Rf from the Bottom Edge of the Plate

Common Trap — — Saying the compound with the highest TLC Rf has the longest column retention time, or dividing by the total length of the glass plate.

Correct Reasoning — — High Rf = travels fast with the mobile phase = SHORTEST column retention time (elutes FIRST)! Always measure both distances from the penciled BASELINE origin to the solvent front.

tlc:app | MCAT Applications

TLC can follow reaction progress by comparing starting-material and product spots under the same conditions; a single spot is not proof of absolute purity. Preparative TLC can isolate material from bands; flash column chromatography uses pressure to drive mobile phase through a packed stationary phase. The analytical TLC trend is not an experimental Rf prediction.

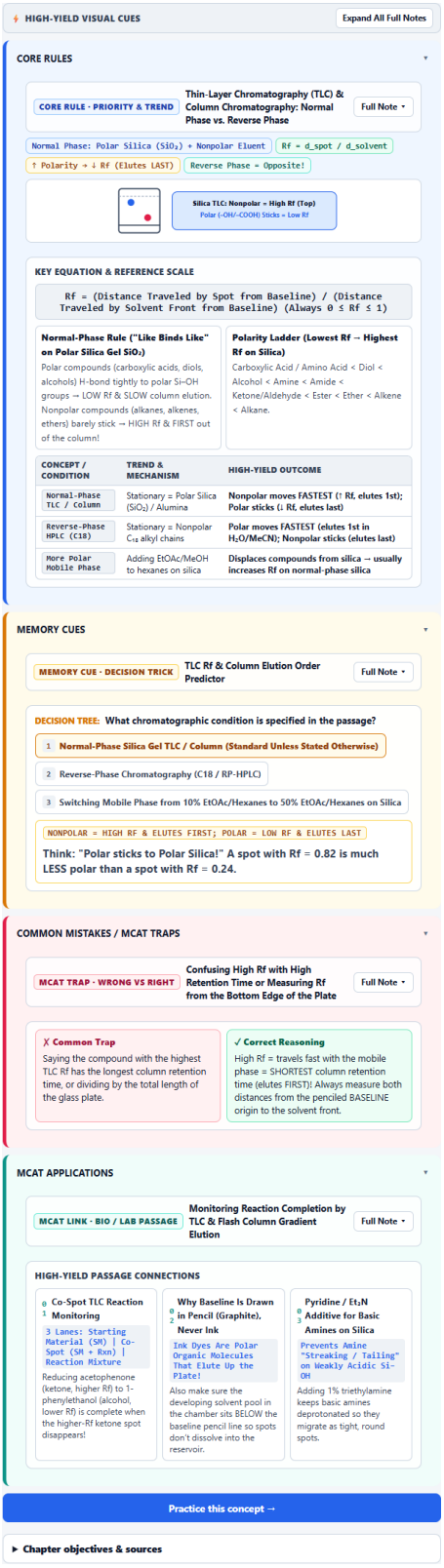
Visual card: Monitoring Reaction Completion by TLC & Flash Column Gradient Elution

Co-Spot TLC Reaction Monitoring — 3 Lanes: Starting Material (SM) | Co-Spot (SM + Rxn) | Reaction Mixture — Reducing acetophenone (ketone, higher Rf) to 1-phenylethanol (alcohol, lower Rf) is complete when the higher-Rf ketone spot disappears!

Why Baseline Is Drawn in Pencil (Graphite), Never Ink — Ink Dyes Are Polar Organic Molecules That Elute Up the Plate! — Also make sure the developing solvent pool in the chamber sits BELOW the baseline pencil line so spots don't dissolve into the reservoir.

Pyridine / Et_3N Additive for Basic Amines on Silica — Prevents Amine "Streaking / Tailing" on Weakly Acidic Si–OH — Adding 1% triethylamine keeps basic amines deprotonated so they migrate as tight, round spots.

Screenshot: tlc-visuals



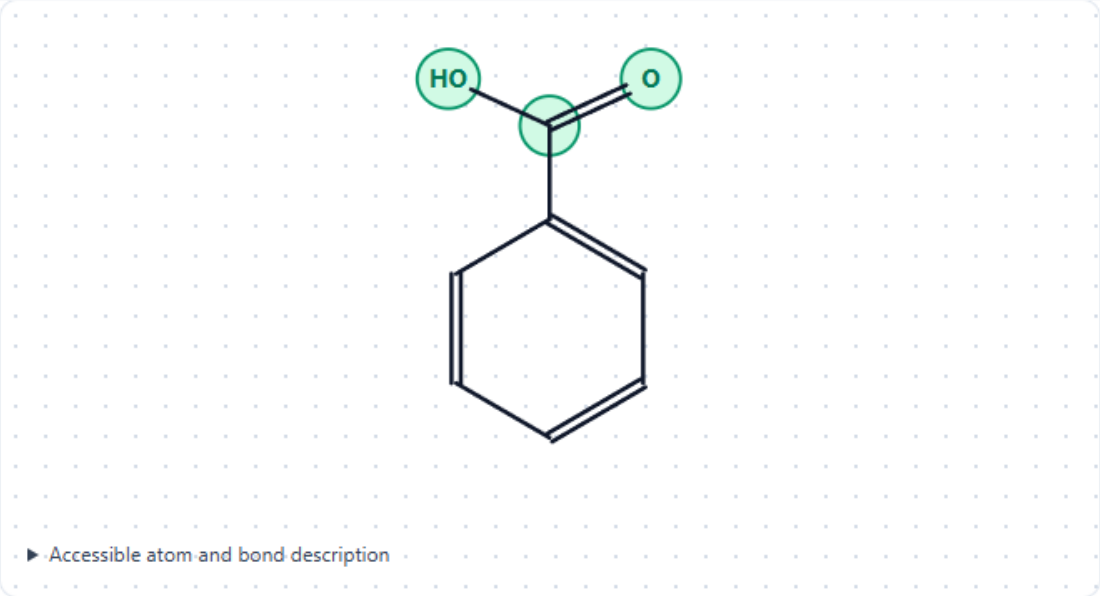
tlc:step-1 | 1. Identify polar interactions

The carboxyl group can interact strongly with silica. Actual retention depends on the eluent, ionization, and other experimental details.

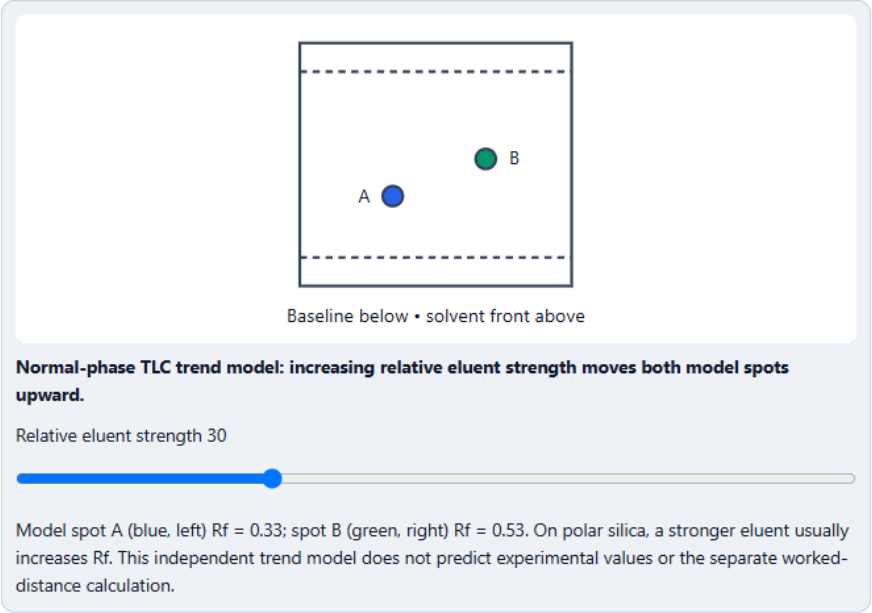
Displayed molecule ID: benzoicacid

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Screenshot: tlc-step-1



Screenshot: tlc-step-1-diagram



{"diagramSpecification": {"type": "tlc", "caption": "Normal-phase TLC trend model: increasing relative eluent strength moves both model spots upward."}}

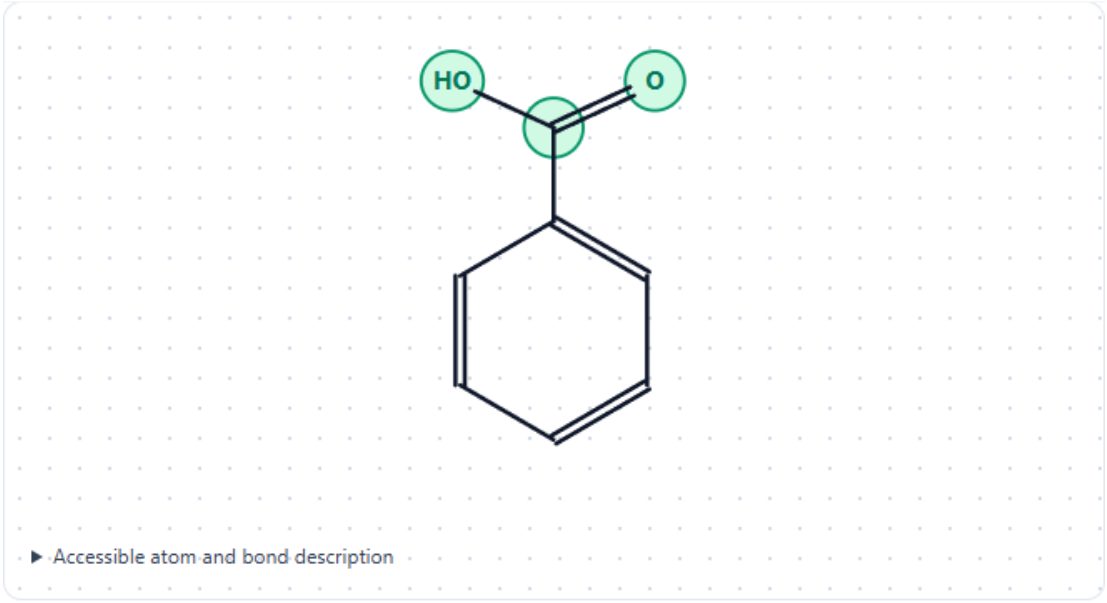
tlc:step-2 | 2. Calculate normalized travel

A spot traveling 1.8 cm while the front travels 6.0 cm has Rf=0.30. Use the same origin for both measurements. The schematic slider is a trend illustration, not a prediction for this molecule.

Displayed molecule ID: benzoicacid

{"highlights": {"prin": [6, 7, 8]}, "revealName": false}

Screenshot: tlc-step-2



specialized | Choose a chromatography method

Match the separation to a molecular property

specialized:rule | Core Rules

Ion exchange separates by charge: a positively charged anion exchanger binds anions; a negatively charged cation exchanger binds cations. Size exclusion separates by access to pores, with larger species generally eluting first. Affinity chromatography uses specific binding. GC requires adequate volatility and thermal stability; HPLC uses a liquid mobile phase under pressure and can handle many nonvolatile analytes.

Visual card: Biochemical Separations: Size-Exclusion (SEC), Ion-Exchange, Affinity & Gas Chromatography

Chemistry diagram: Size-exclusion cartoon: larger analyte tokens bypass pores and generally elute first. Tokens are not molecular structures.

SEC (Size Exclusion): BIG Elutes First!

Cation Exchange (− Bead): Positives Stick

Anion Exchange (+ Bead): Negatives Stick

Affinity = Specific Ligand

Size-Exclusion (SEC / Gel Filtration) | Porous beads (no chemical binding) | LARGE proteins elute FIRST (bypass pores); small enter pores & elute last

Cation-Exchange (Negatively Charged Bead) | Binds positively charged cations (pH < pI) | Anions (pH > pI) wash straight through; elute bound cations with ↑ [NaCl] or ↑ pH

Anion-Exchange (Positively Charged Bead) | Binds negatively charged anions (pH > pI) | Cations (pH < pI) wash through; elute bound anions with ↑ [NaCl] or ↓ pH

Affinity Chromatography | Immobilized ligand / Ni^{2+} / antibody | Highest specificity; elute with free ligand or imidazole (His-tag)

Size-Exclusion (Gel Filtration) Chromatography — — Porous beads act as a molecular maze: LARGE proteins cannot enter the tiny pores and elute FIRST in the void volume; SMALL molecules enter every pore, travel a longer path, and elute LAST!

Ion-Exchange Chromatography (Named for the Ion the Column BINDS!) — — CATION-exchange resin has NEGATIVE beads ($–SO_3^-$) that trap POSITIVE proteins (where pH < pI). ANION-exchange resin has POSITIVE beads ($–NR_3^+$) that trap NEGATIVE proteins (where pH > pI).

Affinity & Chiral Chromatography — — Affinity columns bind one specific target (His_6-tag on Ni^{2+}-NTA; antibody on antigen; biotin on avidin). Chiral stationary phases separate enantiomers without prior derivatization.

specialized:cue | Memory Cues

Ion exchange: charge. Size exclusion: pore access. Affinity: specific interaction.

Visual card: Which Biochemical Column Separates Your Protein Mixture?

What protein property or column type is described in the passage?

Size-Exclusion (SEC) vs. SDS-PAGE Gel Electrophoresis — SEC: Largest Elutes FIRST! | SDS-PAGE: Smallest Migrates FASTEST! — Do not mix them up! In SEC, big molecules bypass bead pores (fast); in polyacrylamide gels, all proteins are forced THROUGH the mesh so small ones win.

Protein with pI = 8.5 in pH 7.0 Buffer: Cation vs. Anion Exchange? — At pH 7.0 < pI 8.5, Protein Is POSITIVE (+): Binds CATION-Exchange (Negative Beads)! — Rule: pH < pI = Protonated (+) = Binds Cation Exchanger; pH > pI = Deprotonated (−) = Binds Anion Exchanger!

How to Elute a Protein Bound to an Affinity or Ion-Exchange Column? — Ion-Exchange: Add High Salt (1M NaCl) or Shift pH | Affinity: Add Free Competitive Ligand! — For a His_6-tagged protein bound to a Ni^{2+}-NTA column, wash away contaminants and elute with free IMIDAZOLE (which mimics the histidine side chain)!

specialized:trap | Common Mistakes / MCAT Traps

Small molecules enter more pore volume and usually elute later in size exclusion. "Anion exchanger" names the species it binds, not the resin's own charge. Reversed-phase HPLC usually retains hydrophobic analytes more strongly than polar ones under comparable conditions.

Visual card: Thinking Small Molecules Elute First in Size-Exclusion or Confusing Cation/Anion Exchanger Bead Charge

Common Trap — — Saying a 15 kDa protein elutes before a 150 kDa protein in size-exclusion chromatography, or saying a cation-exchange bead is positively charged.

Correct Reasoning — — In Size-Exclusion (SEC), LARGE molecules skip the pores and elute FIRST! A CATION exchanger exchanges (binds) positive cations, so the stationary bead itself is NEGATIVELY charged!

specialized:app | MCAT Applications

Protein purification often combines orthogonal methods: charge, size, and affinity. Choose based on protein stability as well as separation power. GC–MS couples separation of suitably volatile analytes with ionization and mass-to-charge analysis. Fragmentation depends on ionization mode; molecular-ion and fragment patterns can support identification, but not every peak is the intact molecular mass.

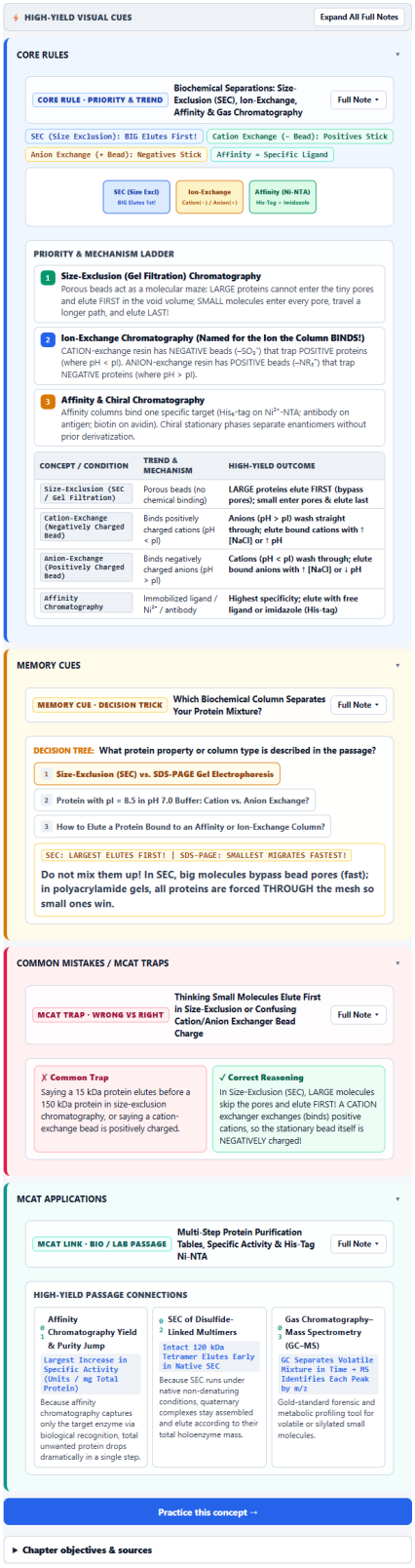
Visual card: Multi-Step Protein Purification Tables, Specific Activity & His-Tag Ni-NTA

Affinity Chromatography Yield & Purity Jump — Largest Increase in Specific Activity (Units / mg Total Protein) — Because affinity chromatography captures only the target enzyme via biological recognition, total unwanted protein drops dramatically in a single step.

SEC of Disulfide-Linked Multimers — Intact 120 kDa Tetramer Elutes Early in Native SEC — Because SEC runs under native non-denaturing conditions, quaternary complexes stay assembled and elute according to their total holoenzyme mass.

Gas Chromatography–Mass Spectrometry (GC–MS) — GC Separates Volatile Mixture in Time → MS Identifies Each Peak by m/z — Gold-standard forensic and metabolic profiling tool for volatile or silylated small molecules.

Screenshot: specialized-visuals



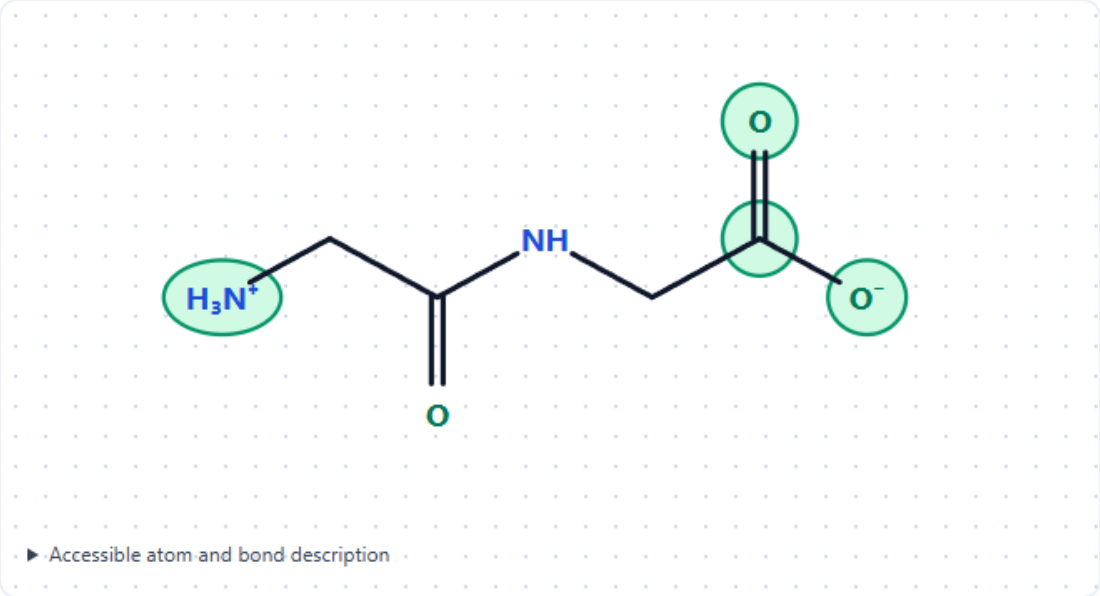
specialized:step-1 | 1. Charge depends on conditions

This peptide has ionizable termini. At pH above its pI it tends toward negative net charge, which can favor binding to a positive anion-exchange resin. Real proteins may have many additional ionizable side chains.

Displayed molecule ID: glycylglycine

{"highlights": {"prin": [0, 6, 7, 8]}, "revealName": true}

Screenshot: specialized-step-1



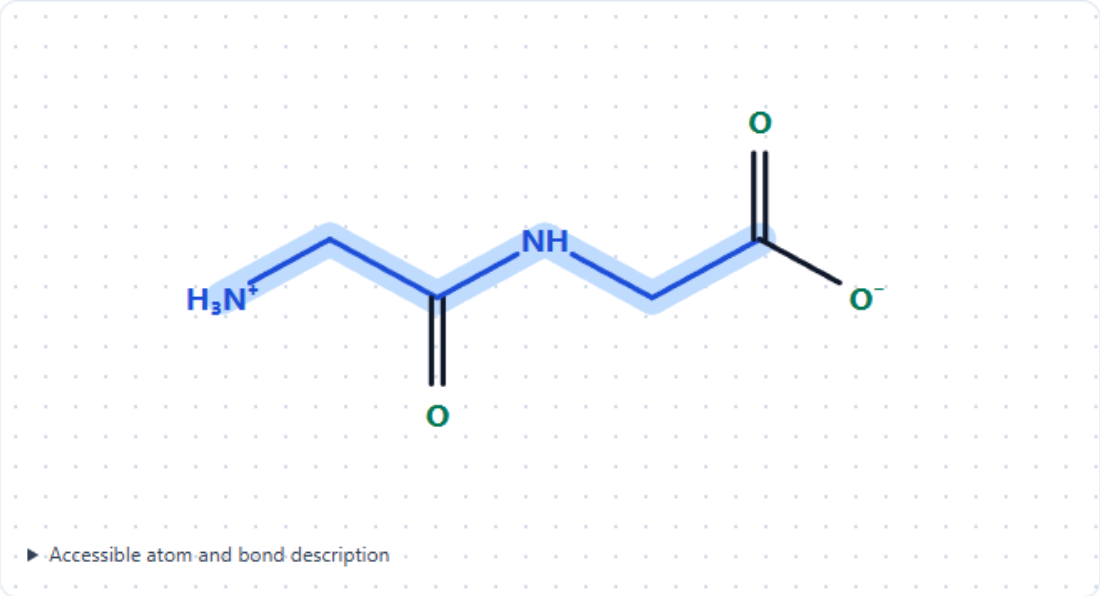
specialized:step-2 | 2. Choose by analyte behavior

An ionic nonvolatile peptide is usually better suited to a liquid-phase method than direct GC without derivatization. Larger proteins can be separated by hydrodynamic size using porous beads; mass alone is not the only determinant.

Displayed molecule ID: glycylglycine

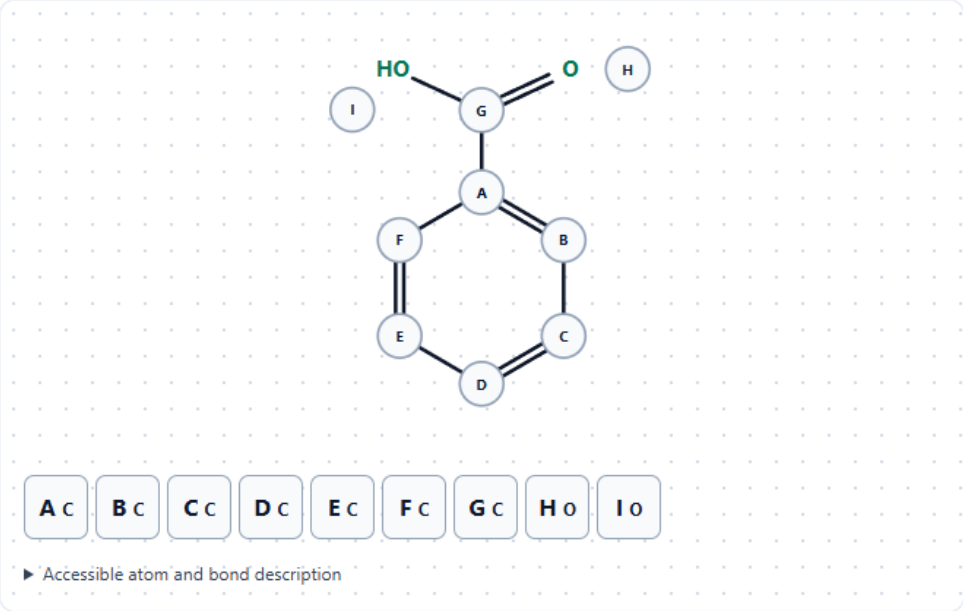
{"highlights": {"path": [0, 1, 2, 4, 5, 6]}, "revealName": false}

Screenshot: specialized-step-2



acid-extraction | Plan an acid–base separation

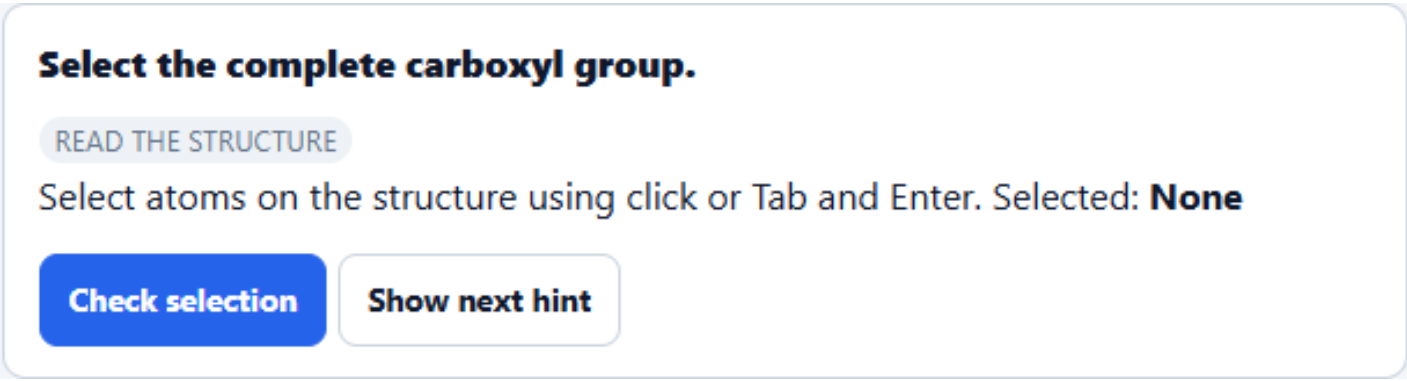
Screenshot: acid-extraction-stage-1-molecule



acid-extraction:stage-1 | Select the complete carboxyl group.

Answer type: atoms

Screenshot: acid-extraction-stage-1-question



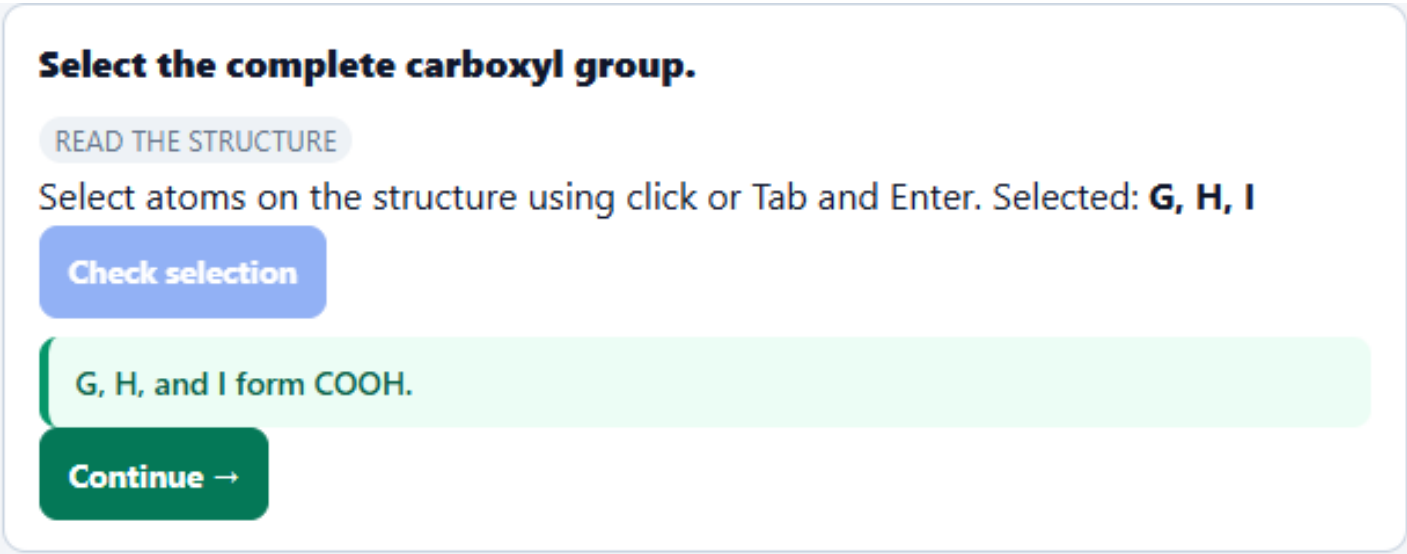
{"authorAnswer": [6, 7, 8], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [6, 7, 8]}}

Hint 1: Find the carbon outside the ring bonded to two O atoms.

Hint 2: Include the carbon and both oxygens.

Explanation: G, H, and I form COOH.

Screenshot: acid-extraction-stage-1-feedback



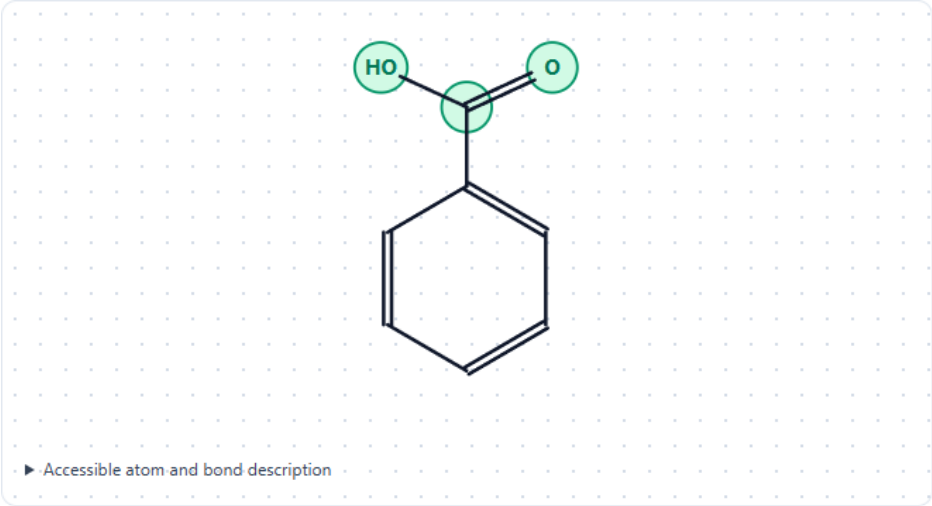
acid-extraction:stage-2 | To favor transfer of this acid into an aqueous phase, choose...

Answer type: choice

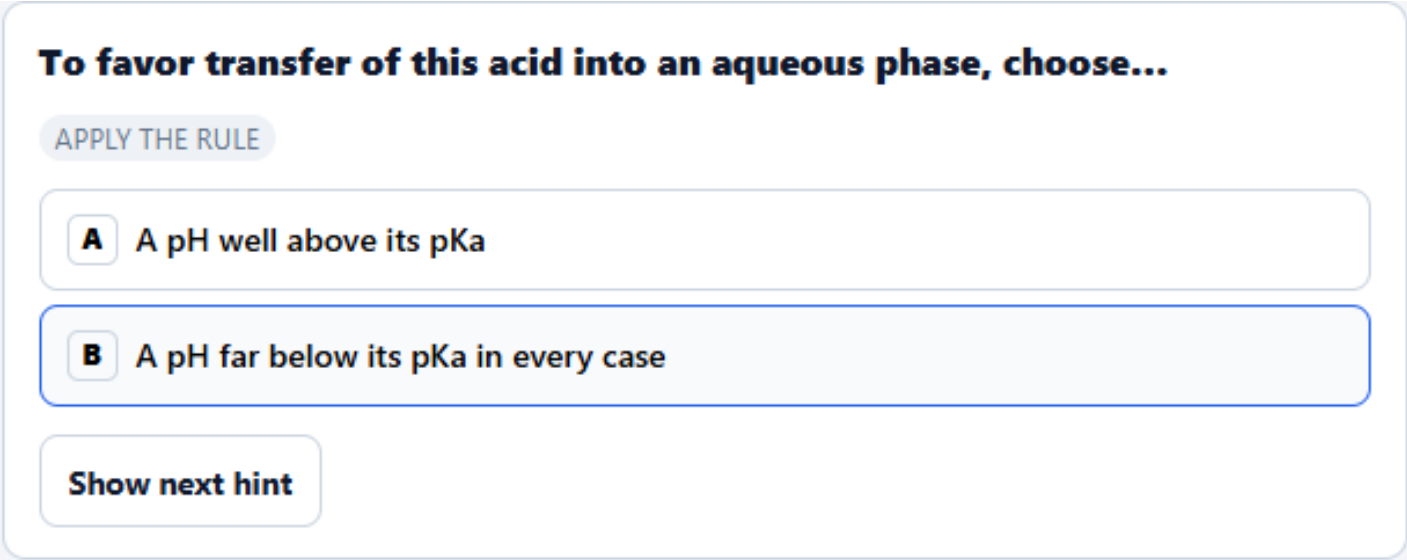
Option 0 (A): A pH well above its pKa

Option 1 (B): A pH far below its pKa in every case

Screenshot: acid-extraction-stage-2-molecule



Screenshot: acid-extraction-stage-2-question



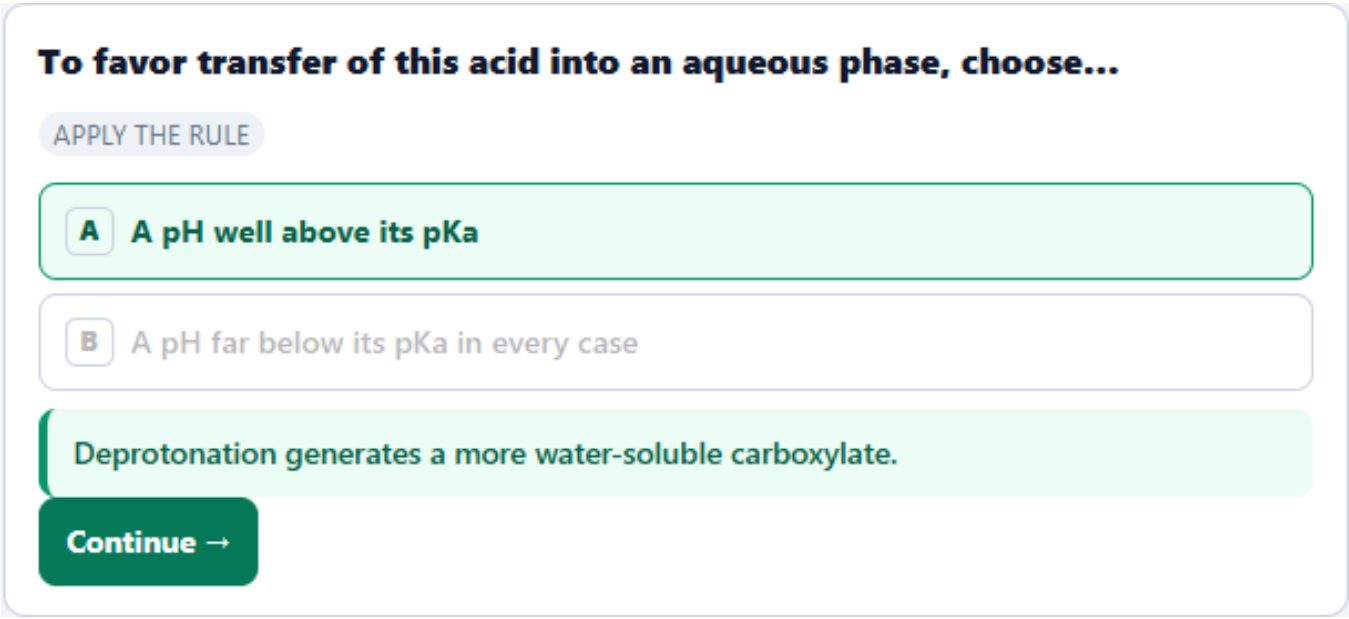
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use ionization to change partitioning.

Hint 2: High pH removes the acidic proton.

Explanation: Deprotonation generates a more water-soluble carboxylate.

Screenshot: acid-extraction-stage-2-feedback



acid-extraction:stage-3 | To collect recrystallized solid from cold mother liquor, use...

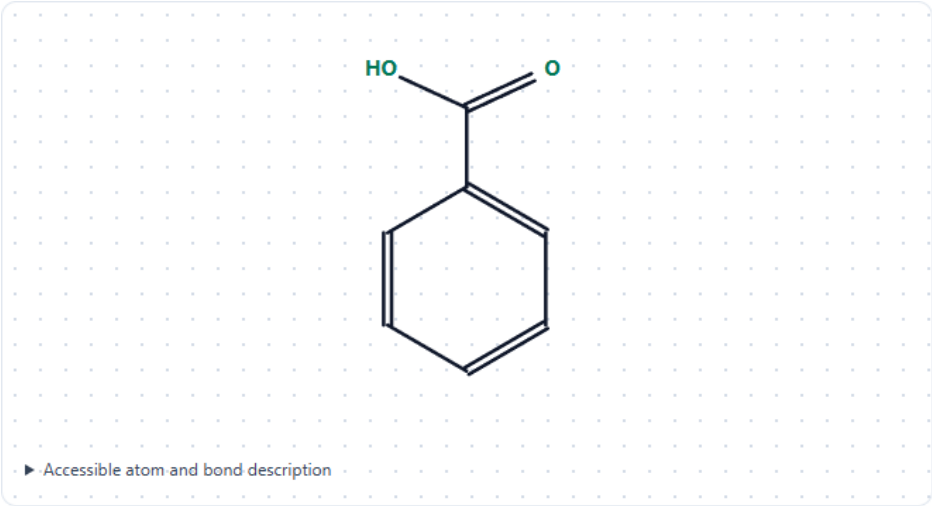
Answer type: choice

Option 0 (A): Vacuum filtration

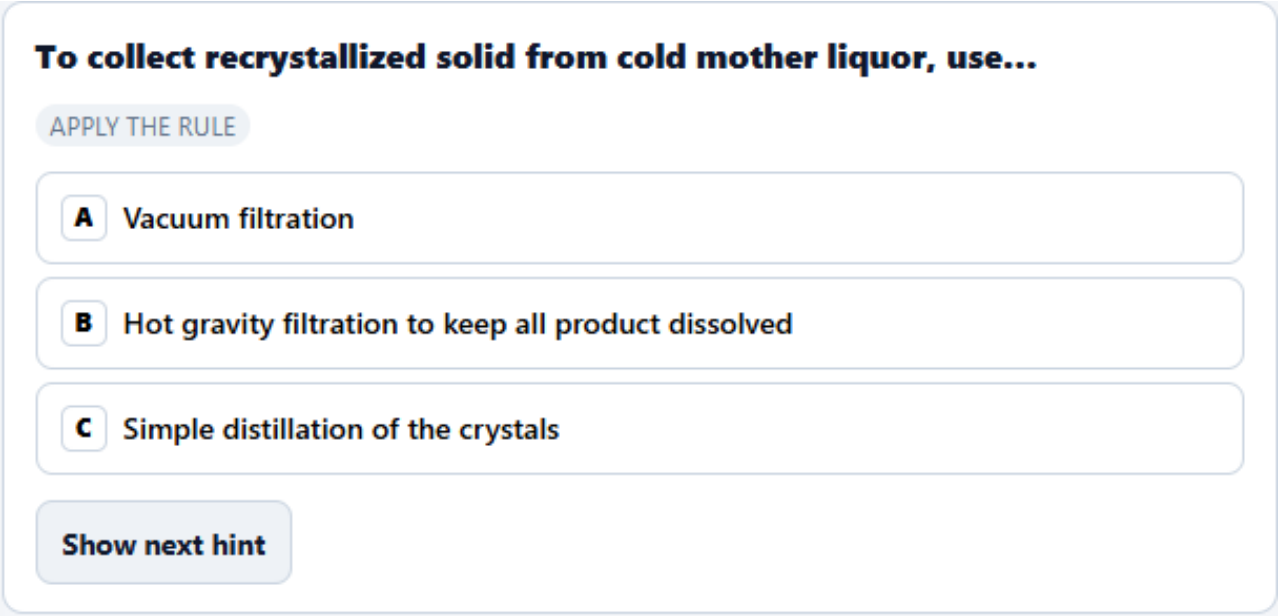
Option 1 (B): Hot gravity filtration to keep all product dissolved

Option 2 (C): Simple distillation of the crystals

Screenshot: acid-extraction-stage-3-molecule



Screenshot: acid-extraction-stage-3-question



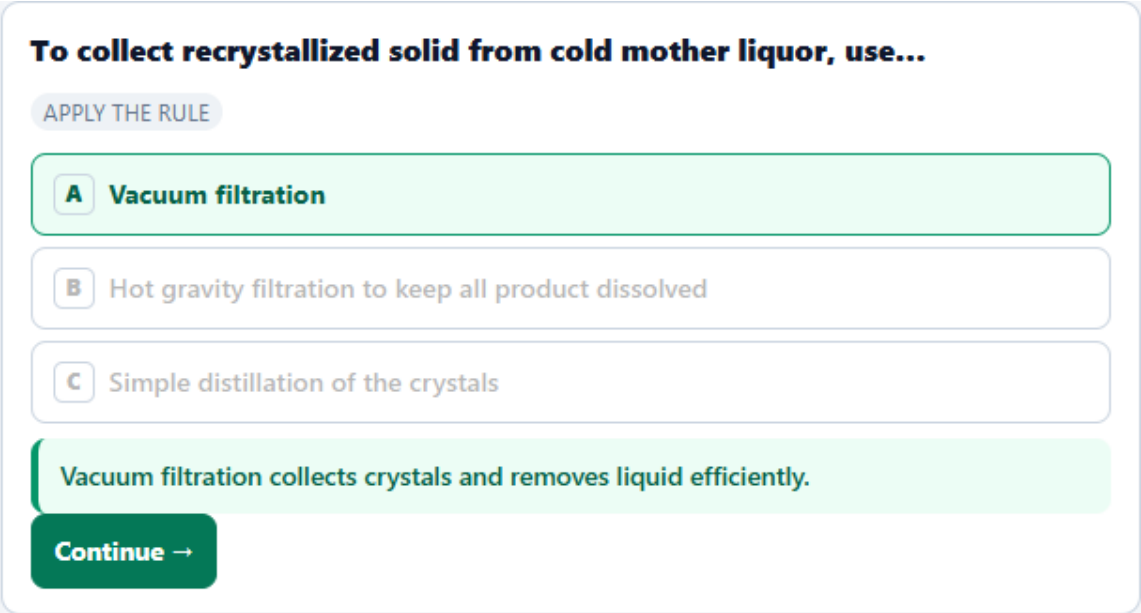
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Identify whether the target is solid or dissolved.

Hint 2: The crystals are the desired residue.

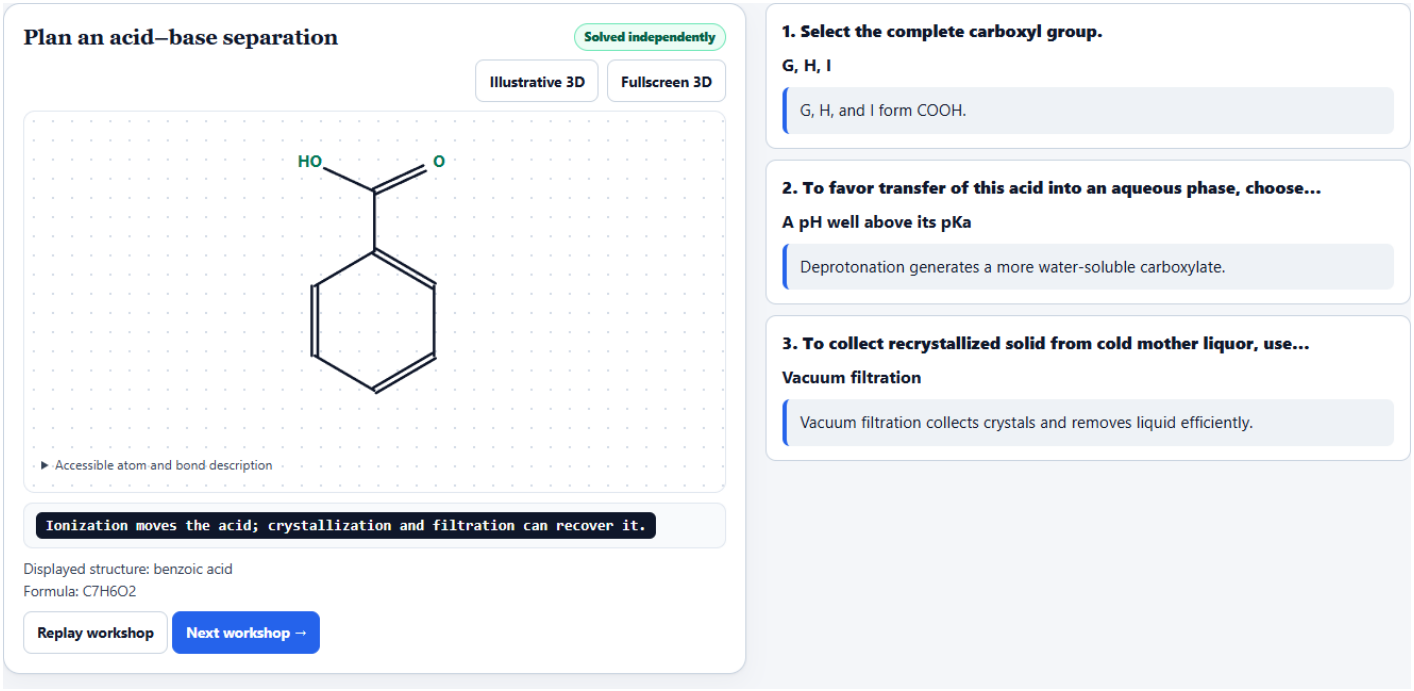
Explanation: Vacuum filtration collects crystals and removes liquid efficiently.

Screenshot: acid-extraction-stage-3-feedback



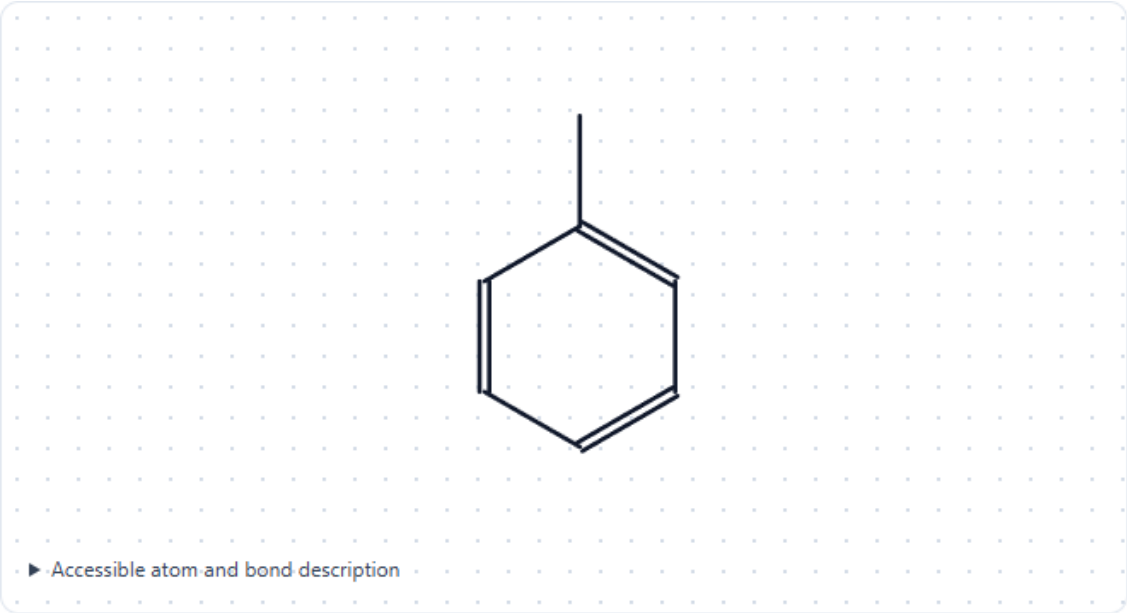
Completion summary: Ionization moves the acid; crystallization and filtration can recover it.

Screenshot: acid-extraction-complete



choose-distillation | Choose the separation conditions

Screenshot: choose-distillation-stage-1-molecule



choose-distillation:stage-1 | A close-boiling liquid mixture is generally better separated with...

Answer type: choice

Option 0 (A): Fractional distillation

Option 1 (B): Simple filtration

Screenshot: choose-distillation-stage-1-question



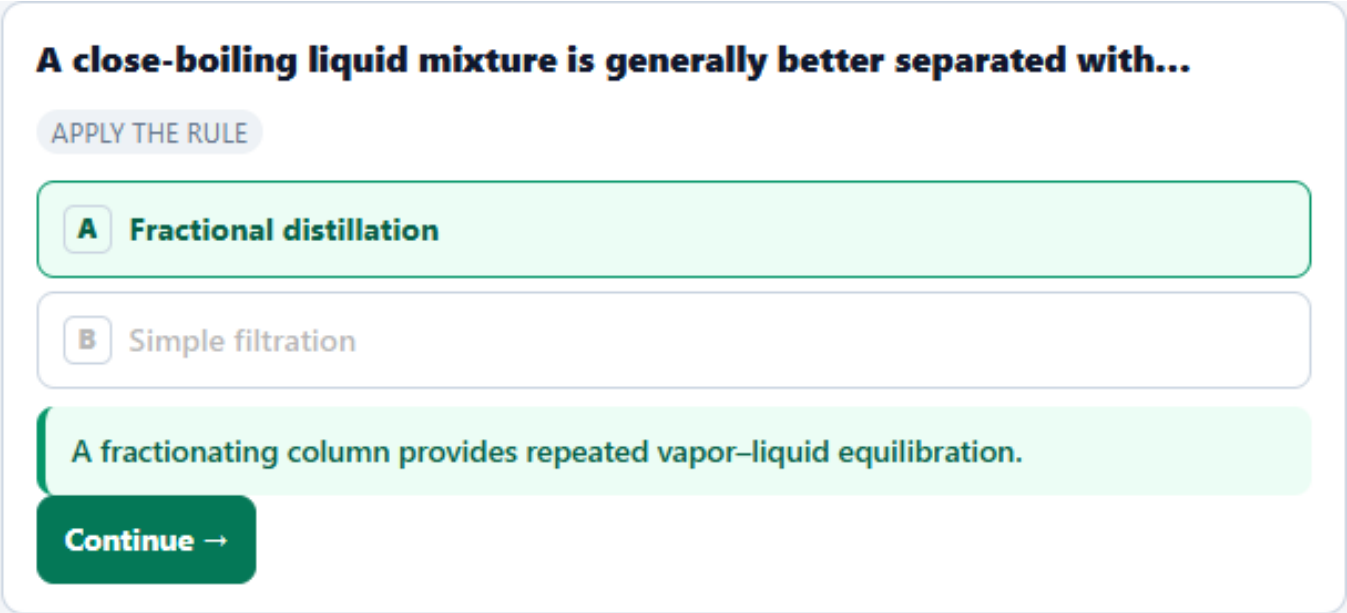
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Both components are liquids.

Hint 2: Repeated enrichment helps when volatility differences are small.

Explanation: A fractionating column provides repeated vapor–liquid equilibration.

Screenshot: choose-distillation-stage-1-feedback



choose-distillation:stage-2 | Reducing external pressure generally makes the boiling temperature...

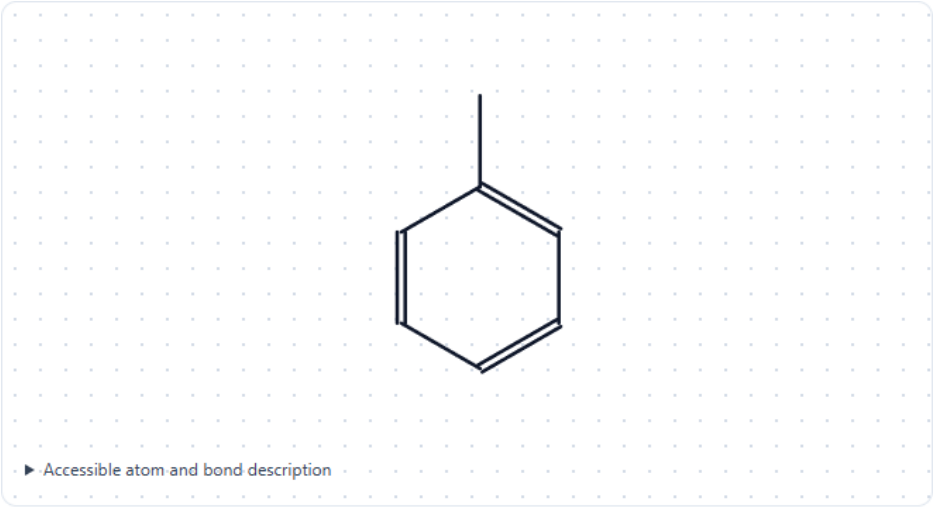
Answer type: choice

Option 0 (A): Lower

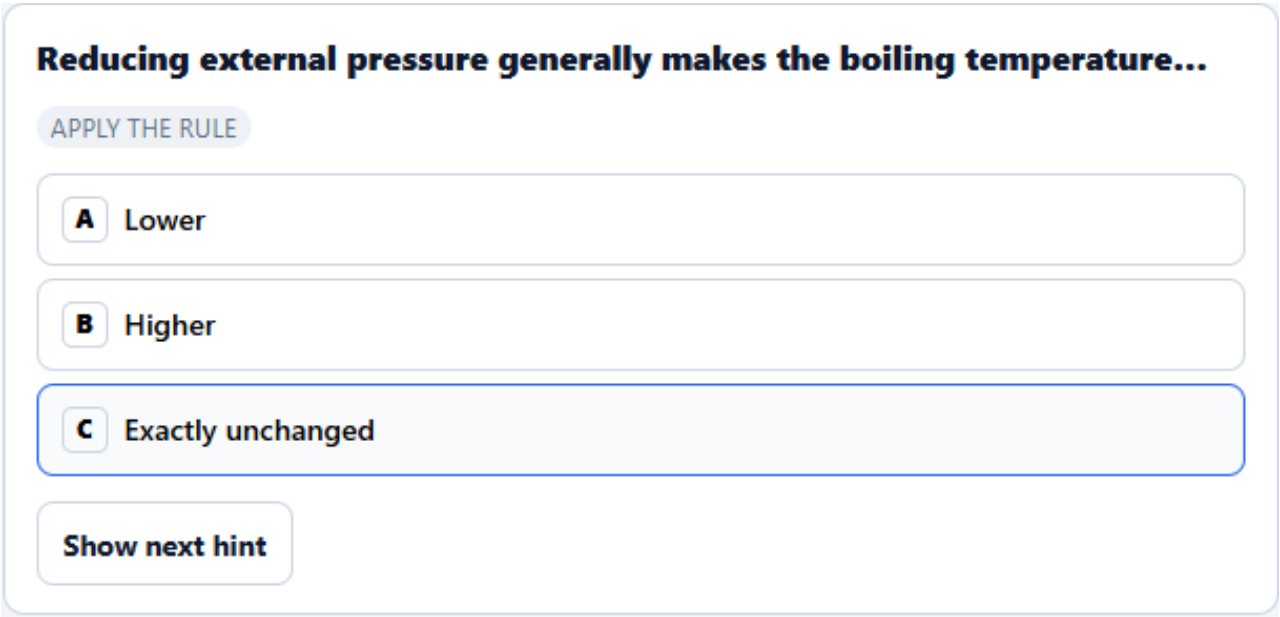
Option 1 (B): Higher

Option 2 (C): Exactly unchanged

Screenshot: choose-distillation-stage-2-molecule



Screenshot: choose-distillation-stage-2-question



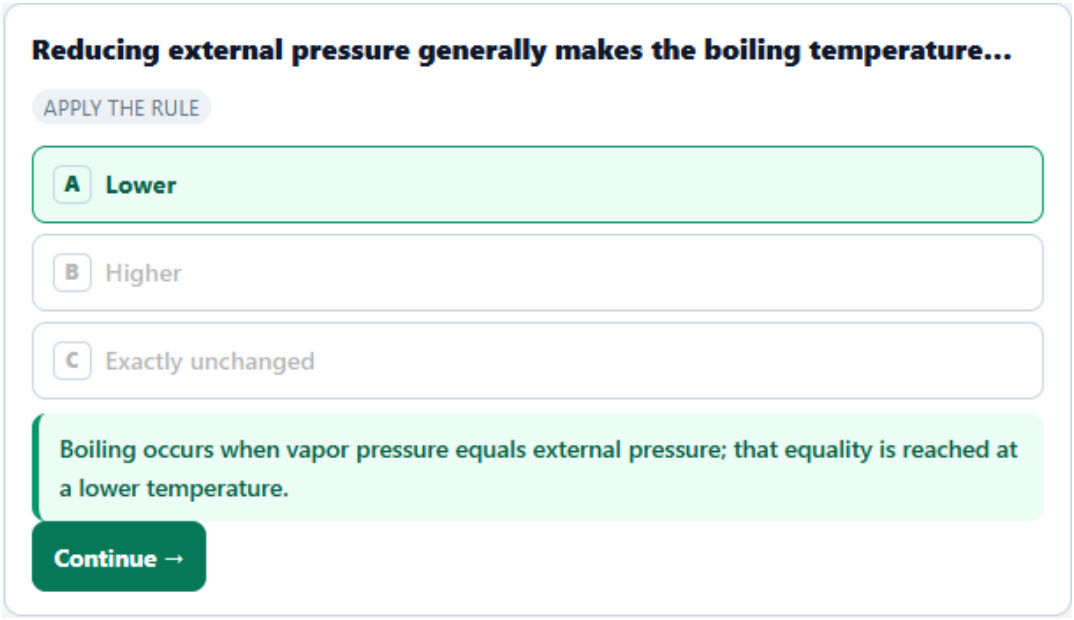
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the condition for boiling.

Hint 2: Less external pressure requires less vapor pressure.

Explanation: Boiling occurs when vapor pressure equals external pressure; that equality is reached at a lower temperature.

Screenshot: choose-distillation-stage-2-feedback



choose-distillation:stage-3 | Which can limit ordinary fractional distillation?

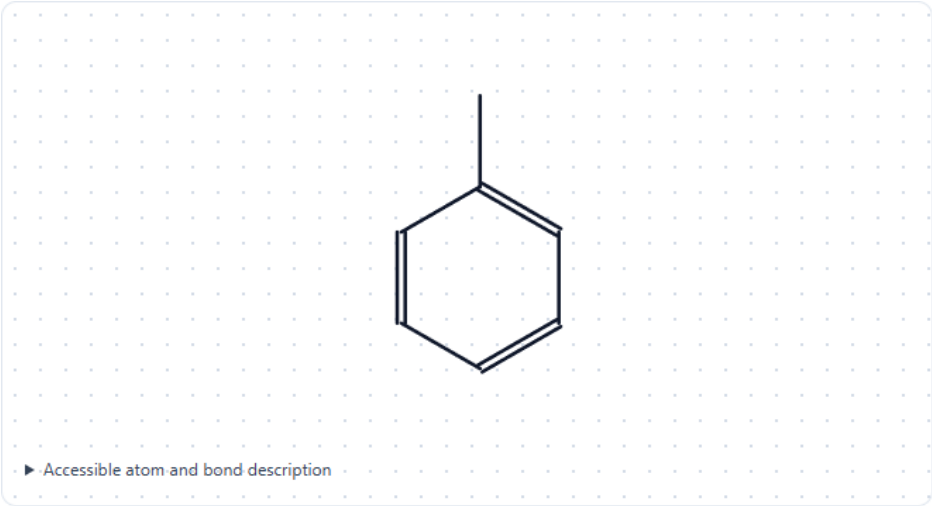
Answer type: choice

Option 0 (A): An azeotrope

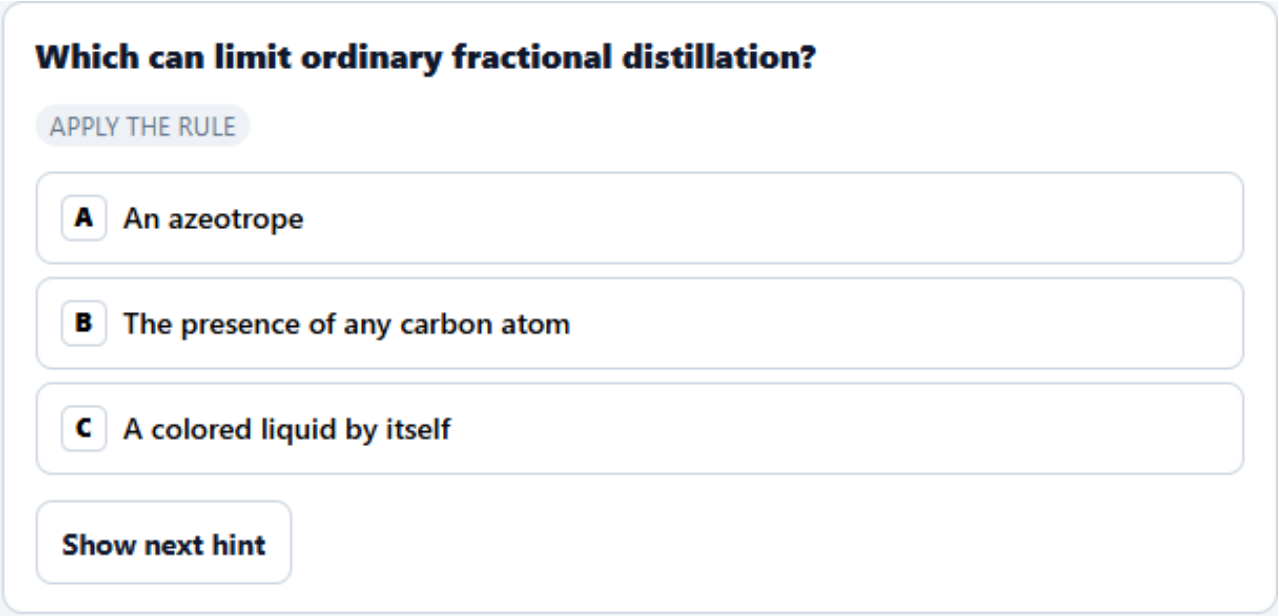
Option 1 (B): The presence of any carbon atom

Option 2 (C): A colored liquid by itself

Screenshot: choose-distillation-stage-3-molecule



Screenshot: choose-distillation-stage-3-question



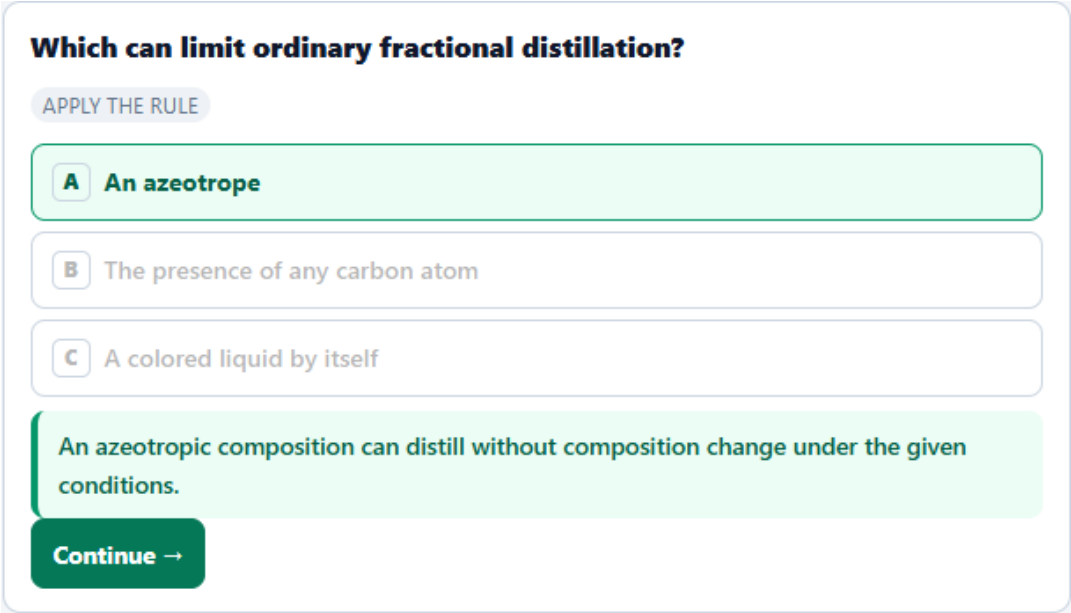
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider nonideal mixture behavior.

Hint 2: A column cannot always surpass the equilibrium limit.

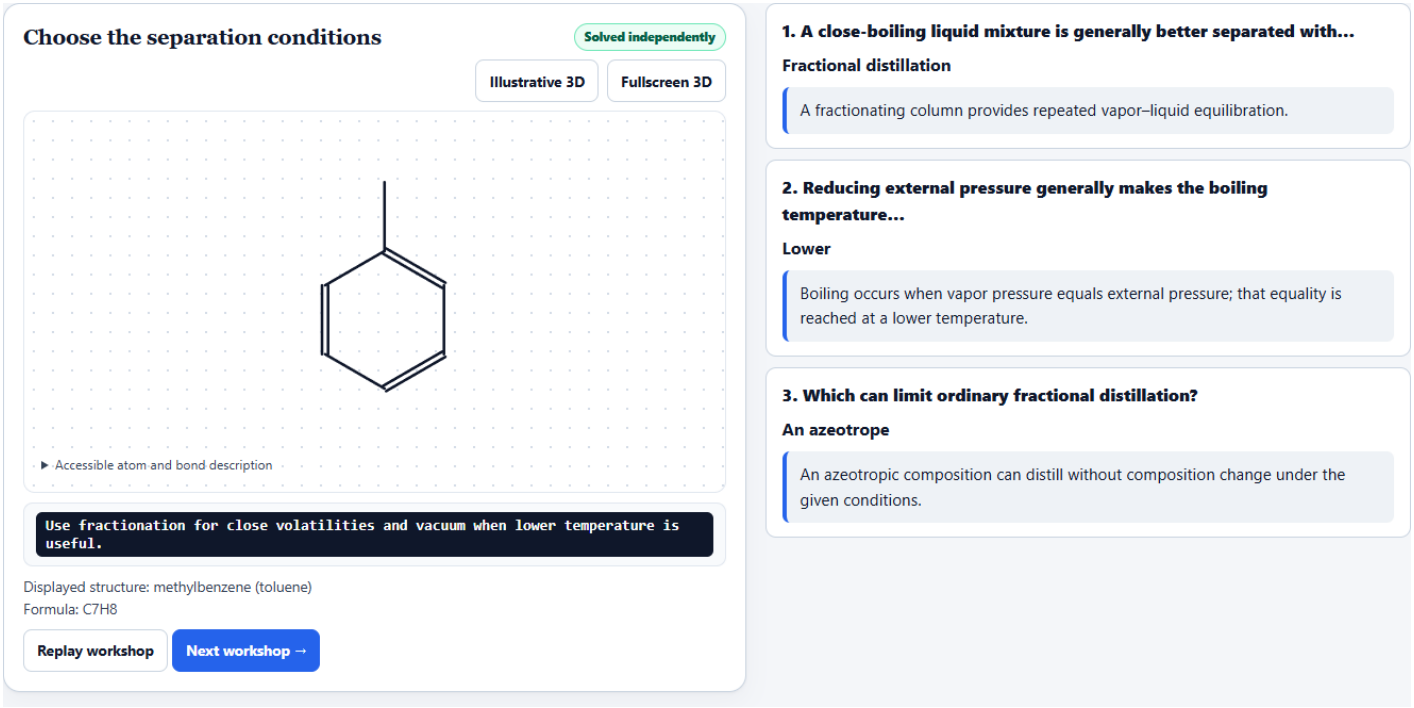
Explanation: An azeotropic composition can distill without composition change under the given conditions.

Screenshot: choose-distillation-stage-3-feedback



Completion summary: Use fractionation for close volatilities and vacuum when lower temperature is useful.

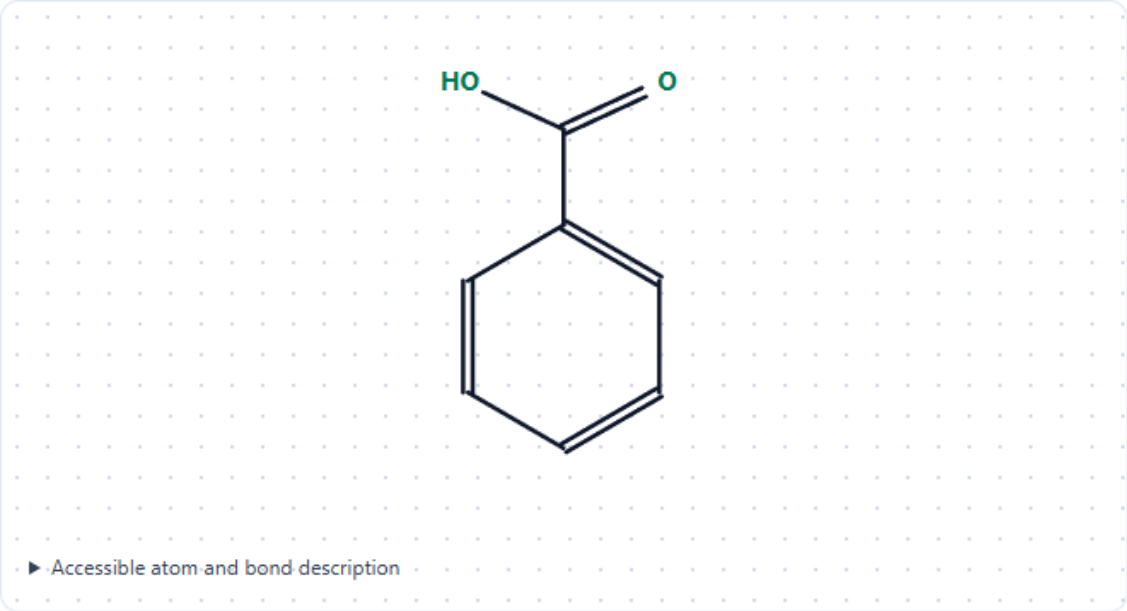
Screenshot: choose-distillation-complete



rf-calculation | Calculate and interpret Rf

Use only the stated travel distances for this calculation. The separate Learn & Explore slider illustrates an independent TLC trend, not these measurements.

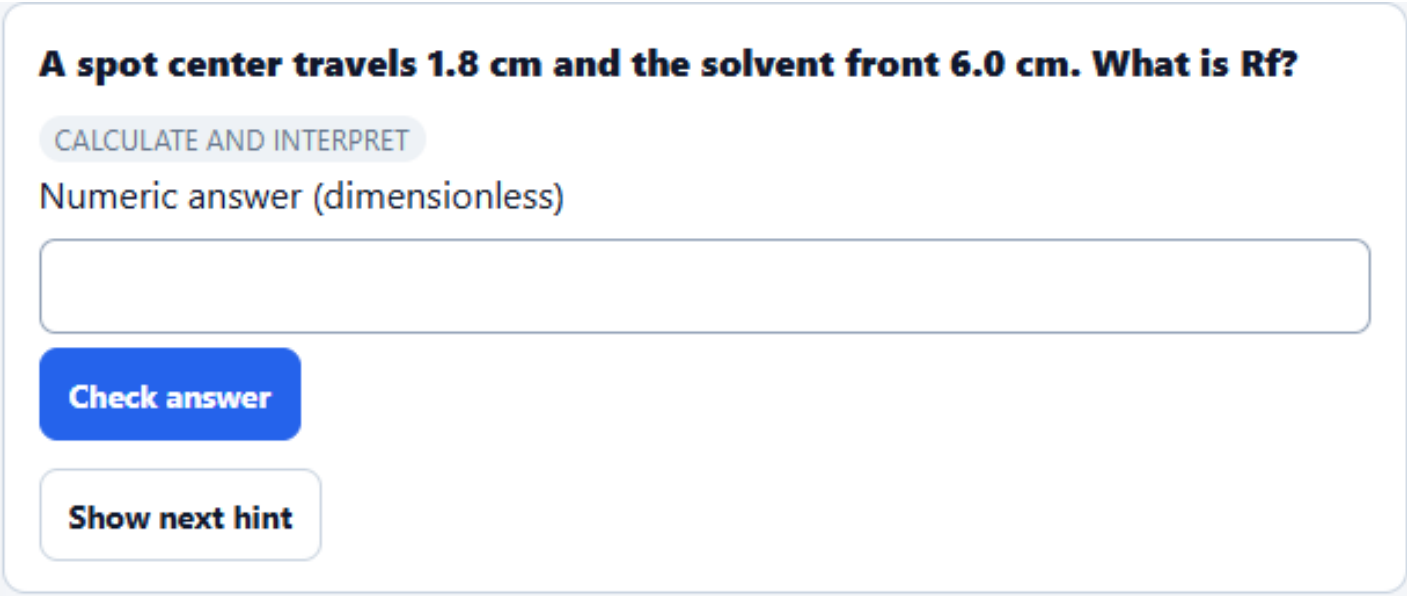
Screenshot: rf-calculation-stage-1-molecule



rf-calculation:stage-1 | A spot center travels 1.8 cm and the solvent front 6.0 cm. What is Rf?

Answer type: number | unit: dimensionless

Screenshot: rf-calculation-stage-1-question



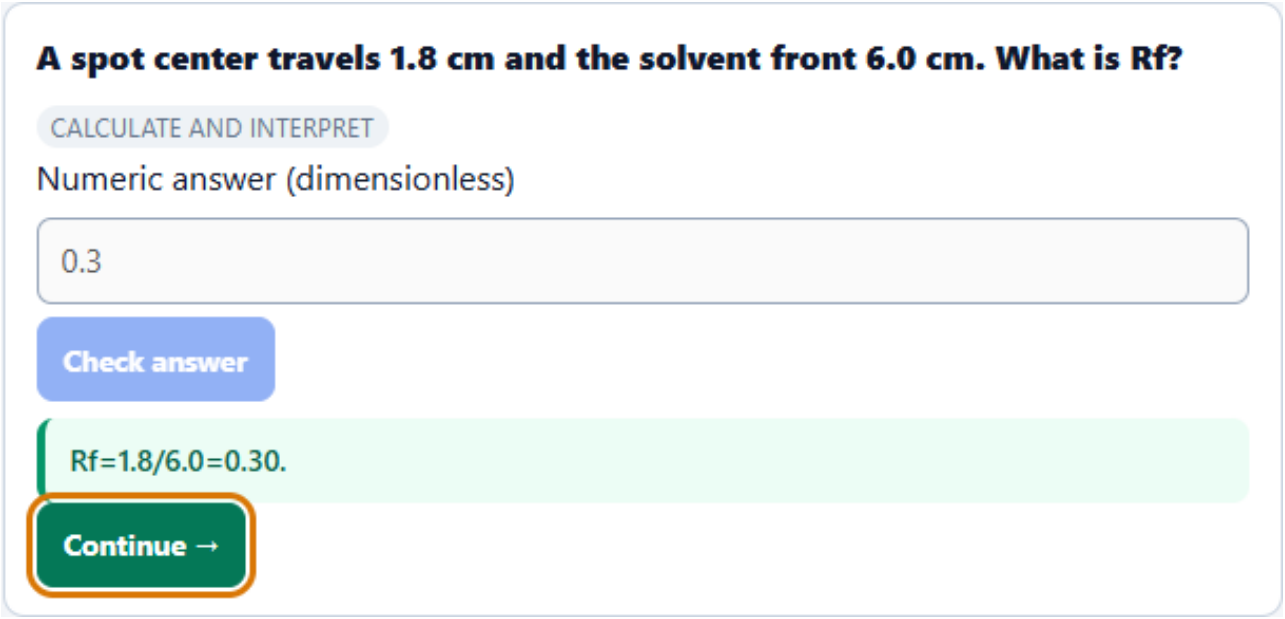
{"authorAnswer": 0.3, "tolerance": 0.005, "highlightsAfterCorrectAnswer": {}}

Hint 1: Divide spot distance by front distance.

Hint 2: Both distances start at the baseline.

Explanation: Rf=1.8/6.0=0.30.

Screenshot: rf-calculation-stage-1-feedback



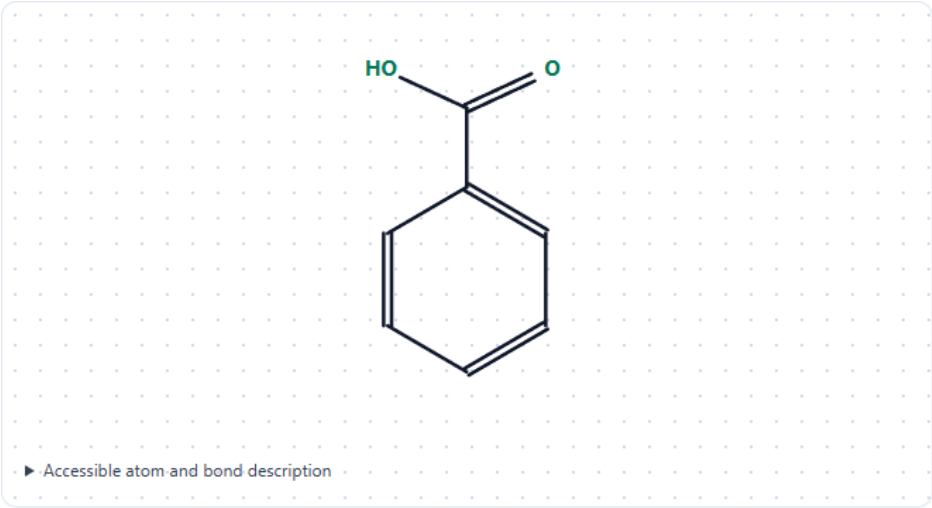
rf-calculation:stage-2 | On normal-phase silica, a stronger eluent usually makes Rf…

Answer type: choice

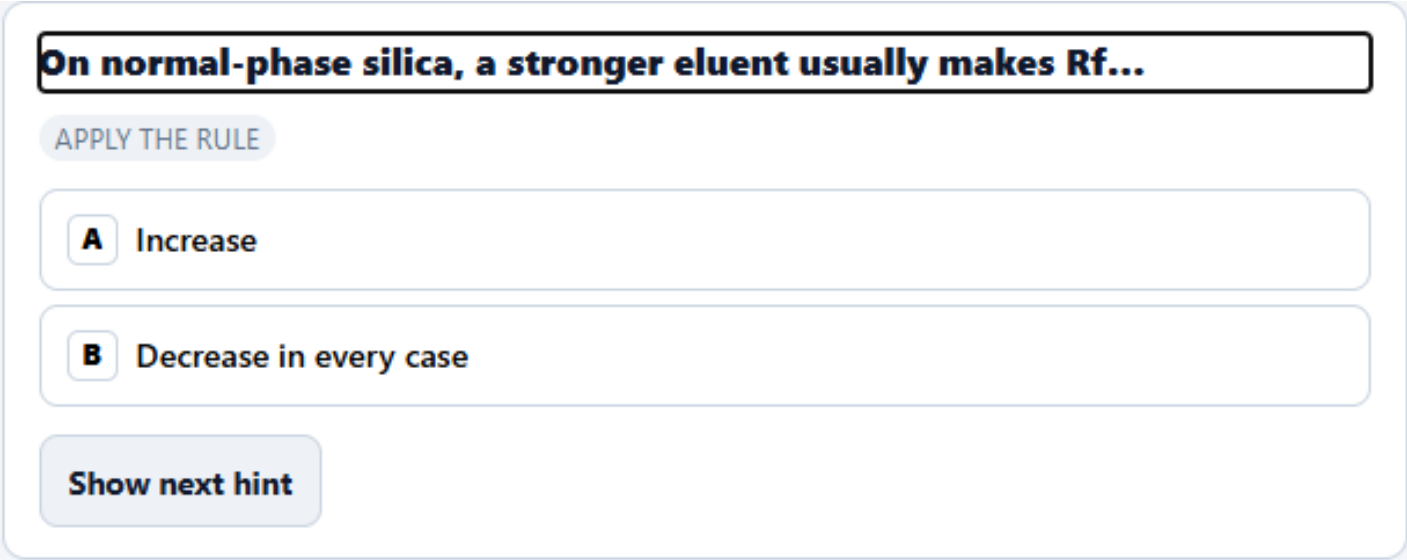
Option 0 (A): Increase

Option 1 (B): Decrease in every case

Screenshot: rf-calculation-stage-2-molecule



Screenshot: rf-calculation-stage-2-question



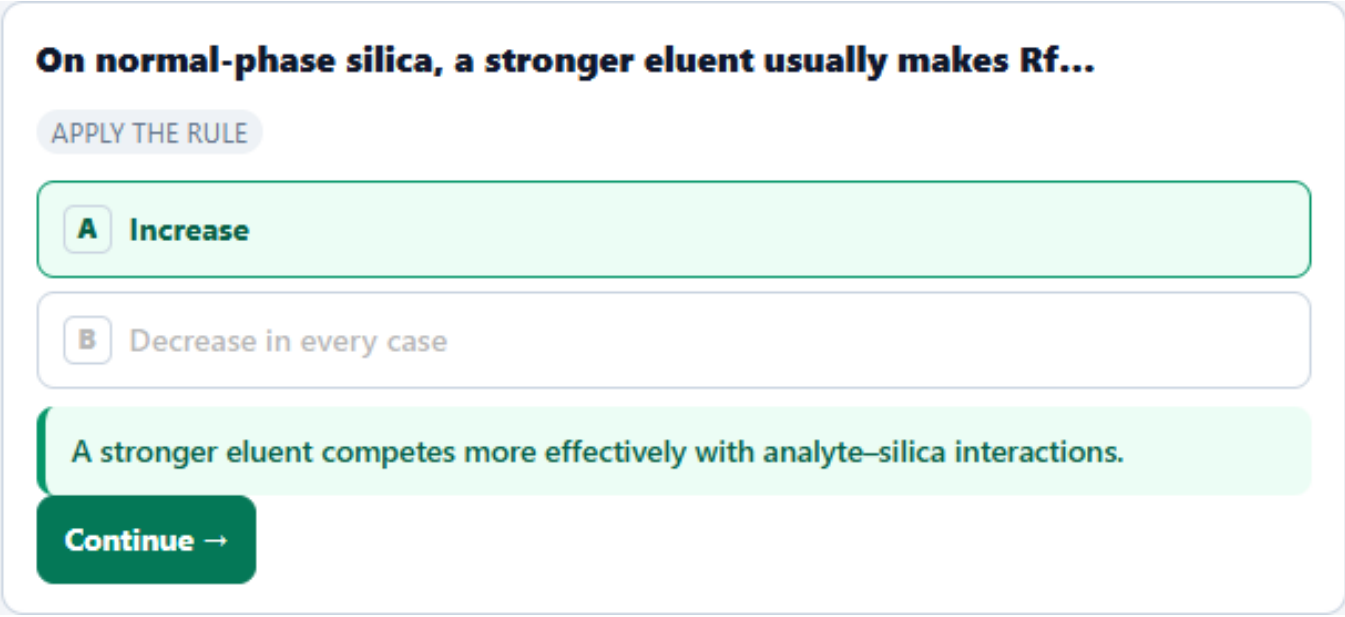
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider the balance between mobile and stationary phase attraction.

Hint 2: More elution usually moves the spot farther.

Explanation: A stronger eluent competes more effectively with analyte–silica interactions.

Screenshot: rf-calculation-stage-2-feedback



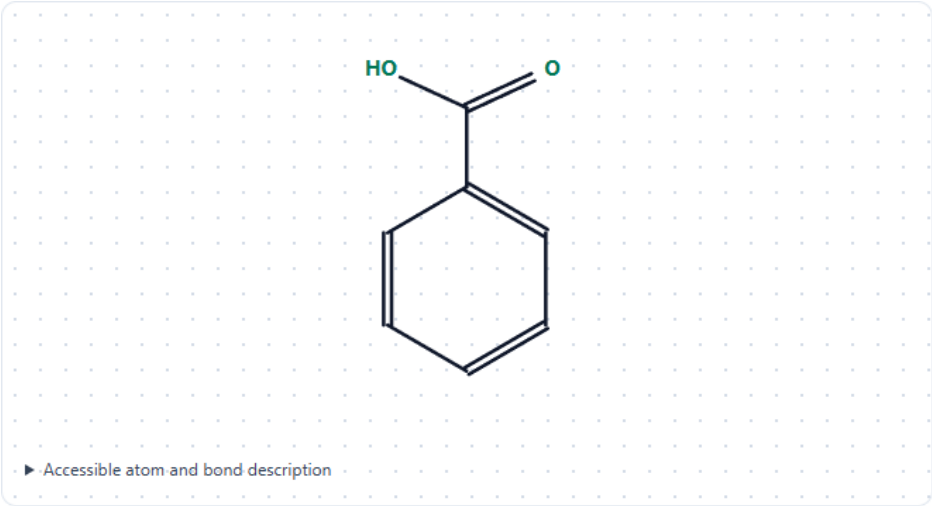
rf-calculation:stage-3 | Two compounds with the same Rf on one plate are...

Answer type: choice

Option 0 (A): Not proven identical

Option 1 (B): Necessarily identical

Screenshot: rf-calculation-stage-3-molecule



Screenshot: rf-calculation-stage-3-question



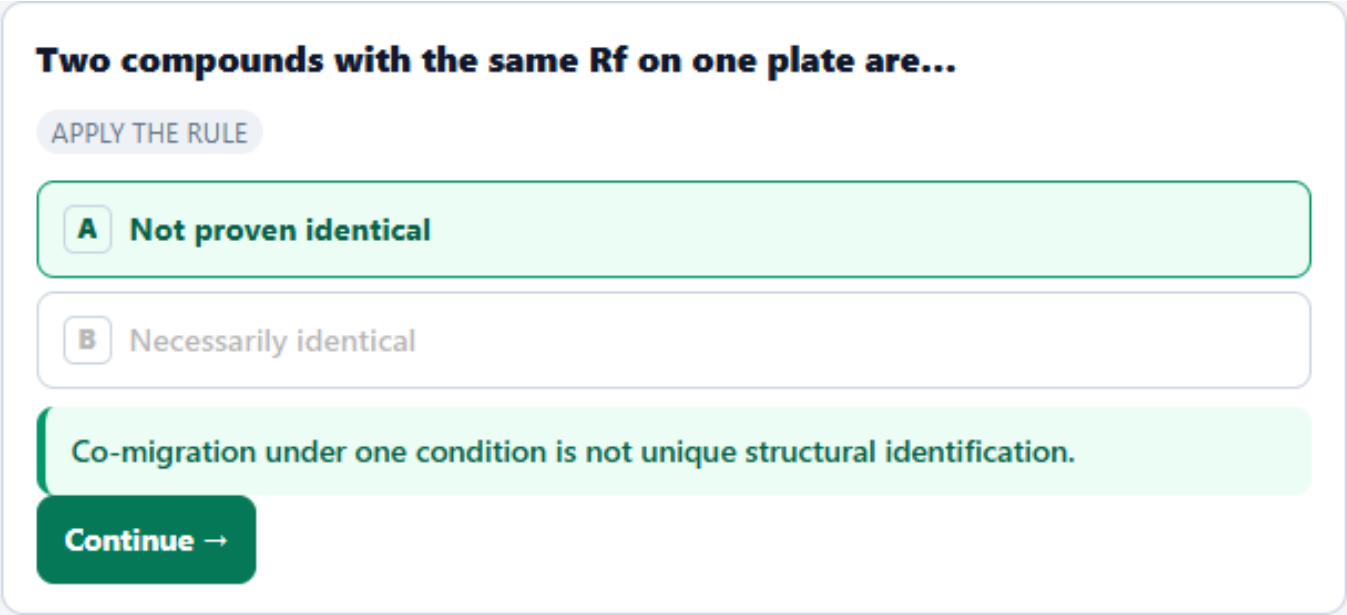
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Different molecules can have similar retention.

Hint 2: Use another condition or independent analytical evidence.

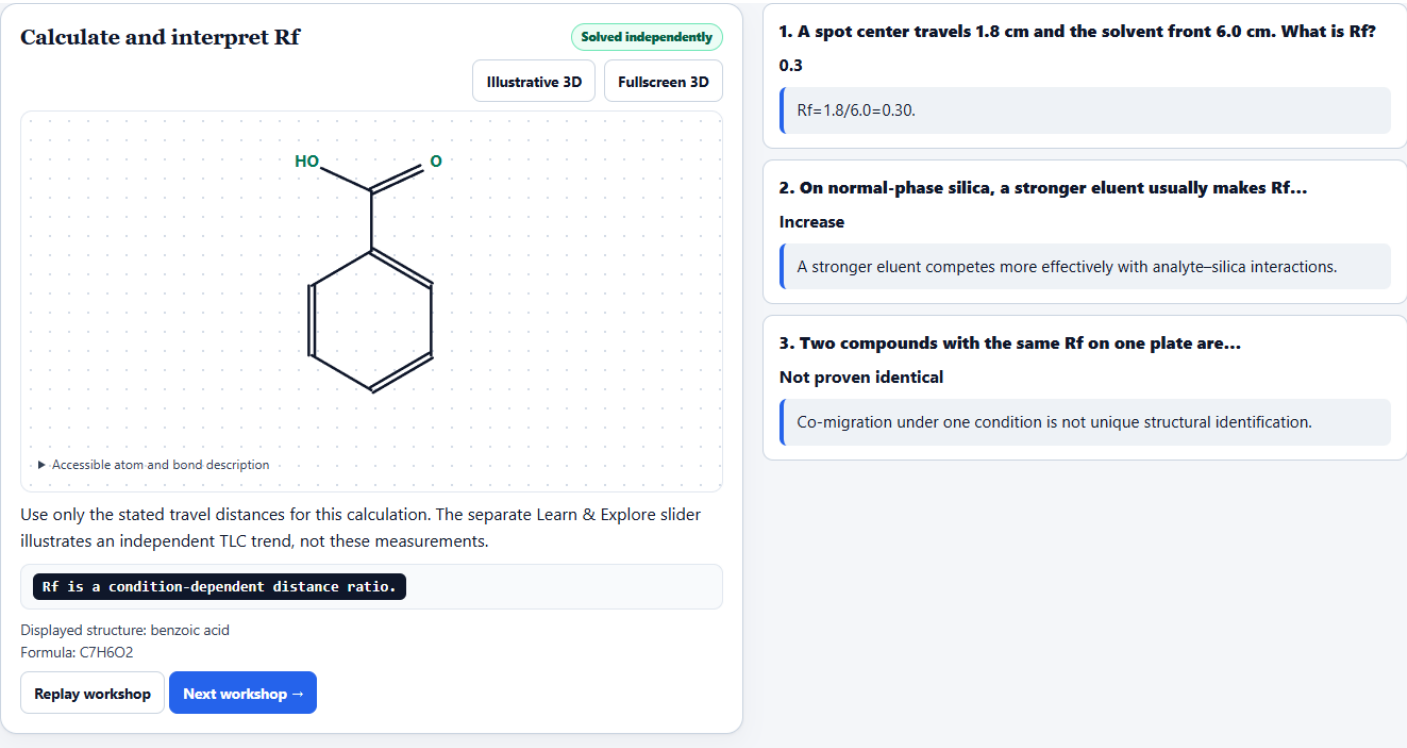
Explanation: Co-migration under one condition is not unique structural identification.

Screenshot: rf-calculation-stage-3-feedback



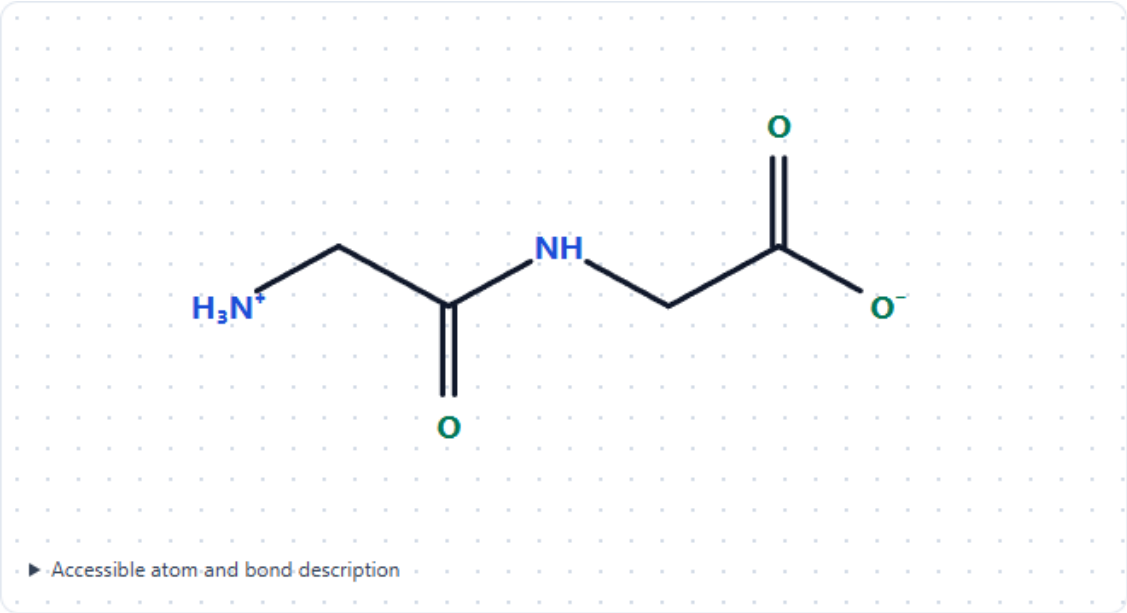
Completion summary: Rf is a condition-dependent distance ratio.

Screenshot: rf-calculation-complete



chromatography-match | Match a protein to a column

Screenshot: chromatography-match-stage-1-molecule



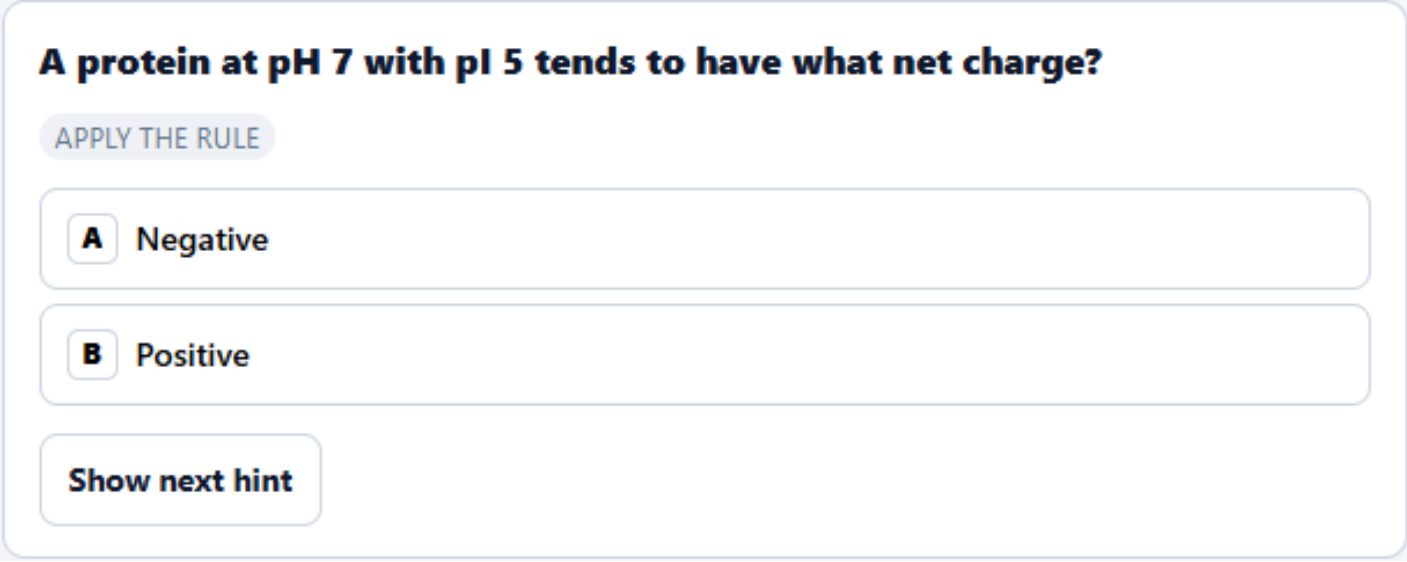
chromatography-match:stage-1 | A protein at pH 7 with pI 5 tends to have what net charge?

Answer type: choice

Option 0 (A): Negative

Option 1 (B): Positive

Screenshot: chromatography-match-stage-1-question



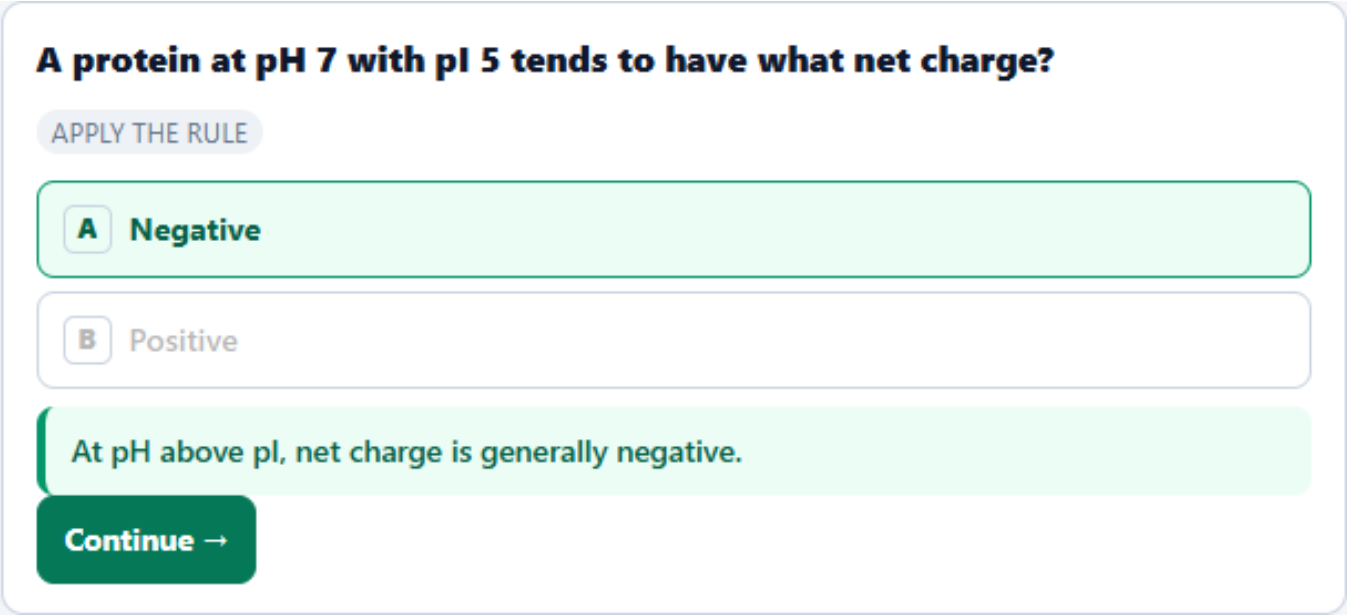
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare pH with pI.

Hint 2: Raising pH removes protons.

Explanation: At pH above pI, net charge is generally negative.

Screenshot: chromatography-match-stage-1-feedback



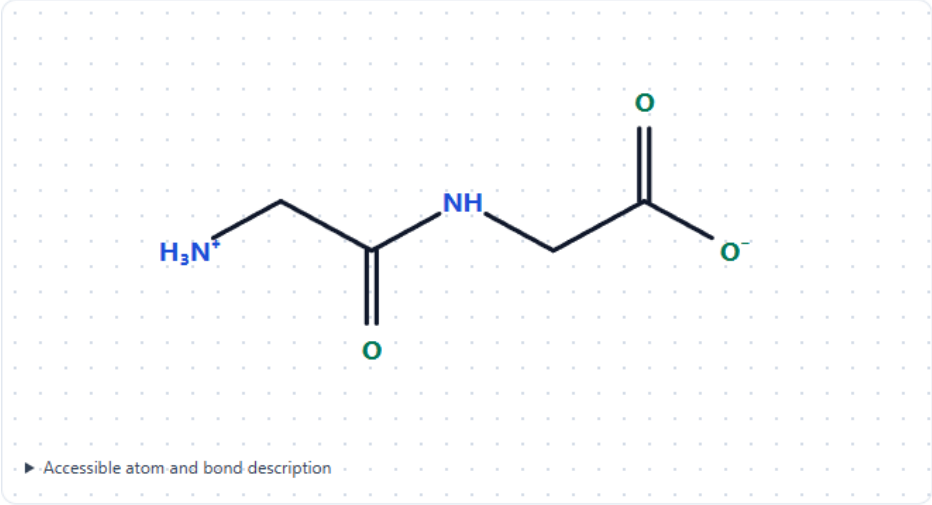
chromatography-match:stage-2 | Which resin would electrostatically bind that protein?

Answer type: choice

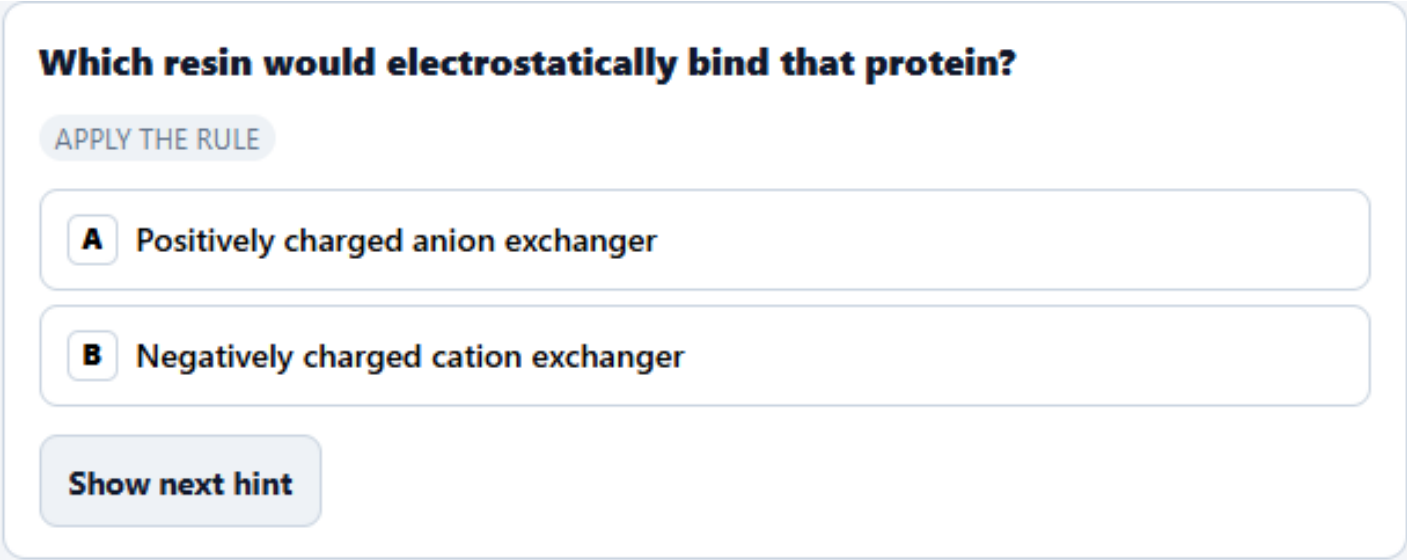
Option 0 (A): Positively charged anion exchanger

Option 1 (B): Negatively charged cation exchanger

Screenshot: chromatography-match-stage-2-molecule



Screenshot: chromatography-match-stage-2-question



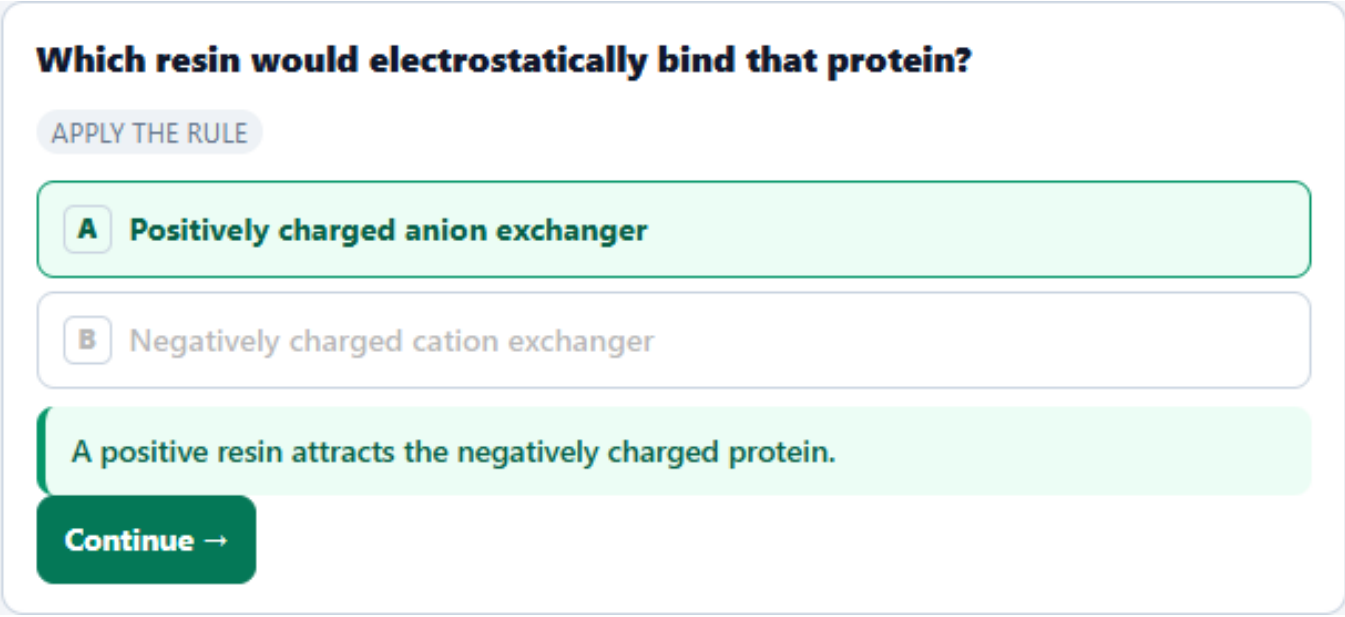
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Opposite charges attract.

Hint 2: Name the exchanger by the ions it binds.

Explanation: A positive resin attracts the negatively charged protein.

Screenshot: chromatography-match-stage-2-feedback



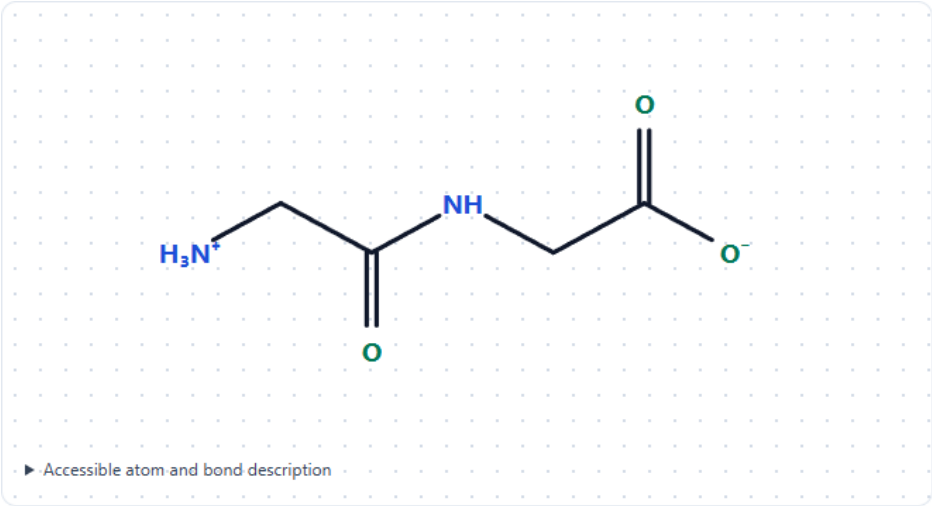
chromatography-match:stage-3 | In ideal size exclusion, larger molecules within the useful separation range usually elute...

Answer type: choice

Option 0 (A): Earlier

Option 1 (B): Later

Screenshot: chromatography-match-stage-3-molecule



Screenshot: chromatography-match-stage-3-question

In ideal size exclusion, larger molecules within the useful separation range usually elute...

APPLY THE RULE

A Earlier

B Later

Show next hint

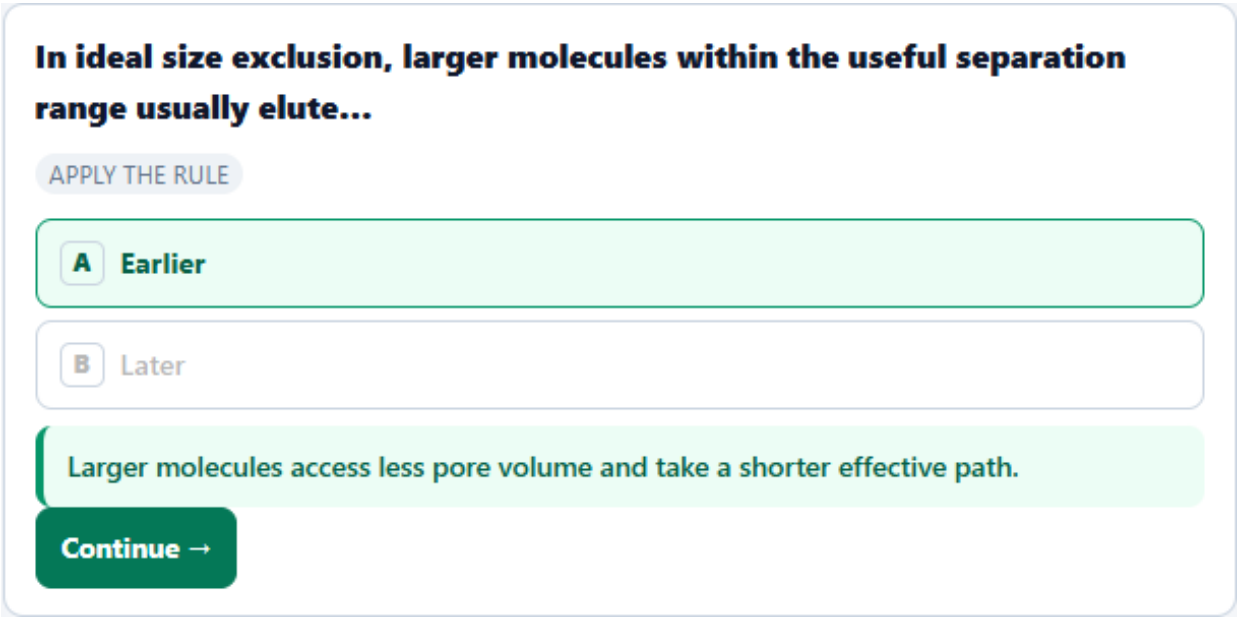
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Smaller molecules explore more pores.

Hint 2: Pore entry delays elution.

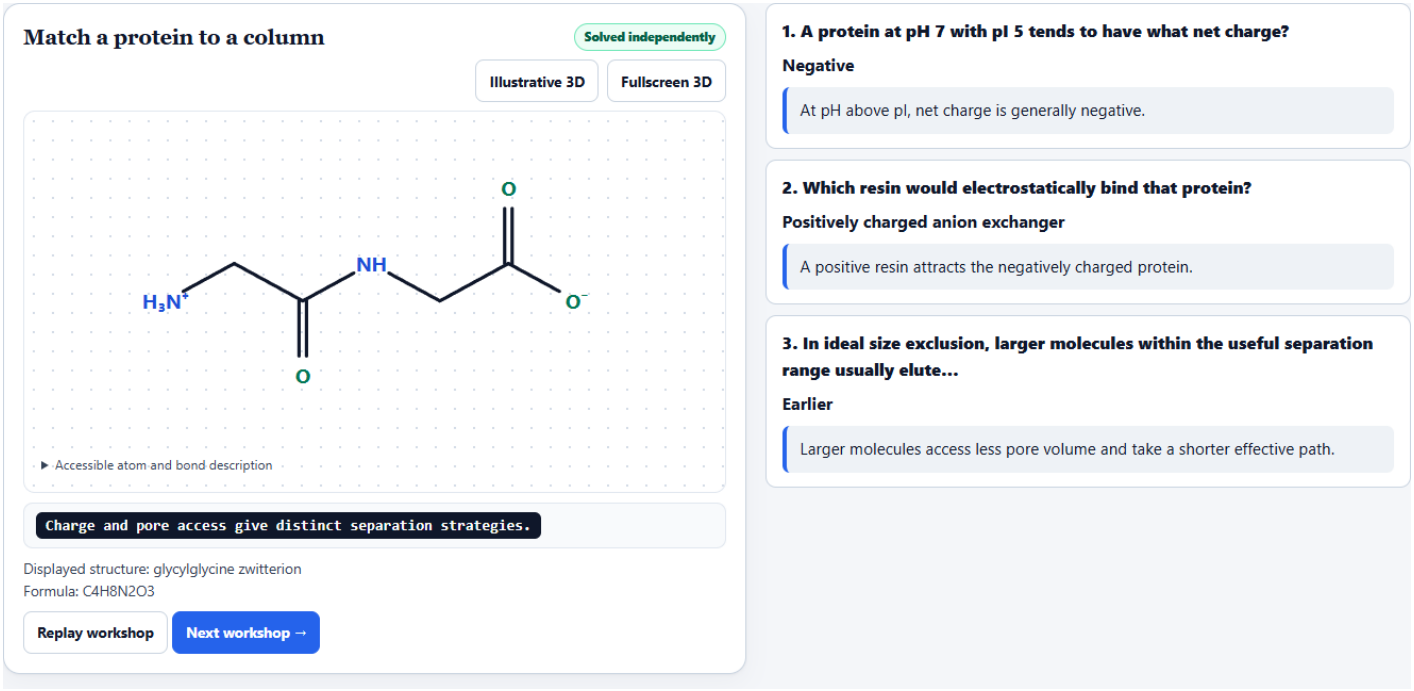
Explanation: Larger molecules access less pore volume and take a shorter effective path.

Screenshot: chromatography-match-stage-3-feedback



Completion summary: Charge and pore access give distinct separation strategies.

Screenshot: chromatography-match-complete



Molecular repository: author claims to verify

benzoicacid

Author claims: benzoic acid | formula C7H6O2 | canonical composition C7H6O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=160.00, y=95.00; charge=0; explicit H override=none

1 (B): C; x=220.00, y=130.00; charge=0; explicit H override=none

2 (C): C; x=220.00, y=200.00; charge=0; explicit H override=none

3 (D): C; x=160.00, y=235.00; charge=0; explicit H override=none

4 (E): C; x=100.00, y=200.00; charge=0; explicit H override=none

5 (F): C; x=100.00, y=130.00; charge=0; explicit H override=none

6 (G): C; x=160.00, y=35.00; charge=0; explicit H override=none

7 (H): O; x=225.00, y=5.00; charge=0; explicit H override=none

8 (I): O; x=95.00, y=5.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [6, 7, 2], [6, 8, 1]]

benzoate

Author claims: benzoate | formula C7H5O2 | canonical composition C7H5O2 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=160.00, y=95.00; charge=0; explicit H override=none

1 (B): C; x=220.00, y=130.00; charge=0; explicit H override=none

2 (C): C; x=220.00, y=200.00; charge=0; explicit H override=none

3 (D): C; x=160.00, y=235.00; charge=0; explicit H override=none

4 (E): C; x=100.00, y=200.00; charge=0; explicit H override=none

5 (F): C; x=100.00, y=130.00; charge=0; explicit H override=none

6 (G): C; x=160.00, y=35.00; charge=0; explicit H override=none

7 (H): O; x=225.00, y=5.00; charge=0; explicit H override=none

8 (I): O; x=95.00, y=5.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [6, 7, 2], [6, 8, 1]]

toluene

Author claims: methylbenzene (toluene) | formula C7H8 | canonical composition C7H8 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=93.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=127.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=195.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=229.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=195.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=127.00; charge=0; explicit H override=none

6 (G): C; x=170.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1]]

glycylglycine

Author claims: glycylglycine zwitterion | formula C4H8N2O3 | canonical composition C4H8N2O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=35.00, y=110.00; charge=1; explicit H override=none

1 (B): C; x=90.00, y=80.00; charge=0; explicit H override=none

2 (C): C; x=145.00, y=110.00; charge=0; explicit H override=none

3 (D): O; x=145.00, y=170.00; charge=0; explicit H override=none

4 (E): N; x=200.00, y=80.00; charge=0; explicit H override=none

5 (F): C; x=255.00, y=110.00; charge=0; explicit H override=none

6 (G): C; x=310.00, y=80.00; charge=0; explicit H override=none

7 (H): O; x=310.00, y=20.00; charge=0; explicit H override=none

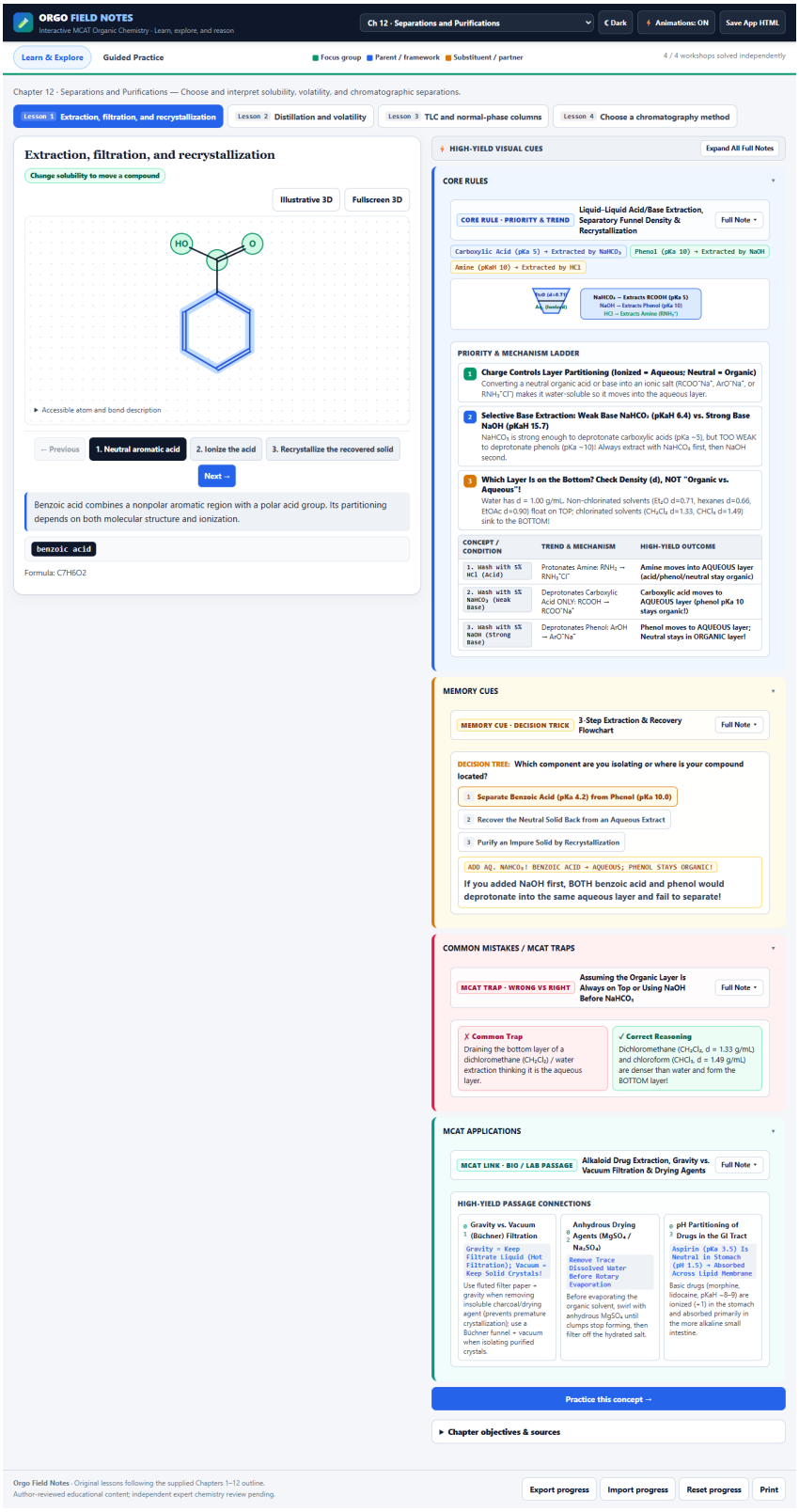
8 (I): O; x=365.00, y=110.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 2], [6, 8, 1]]

Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



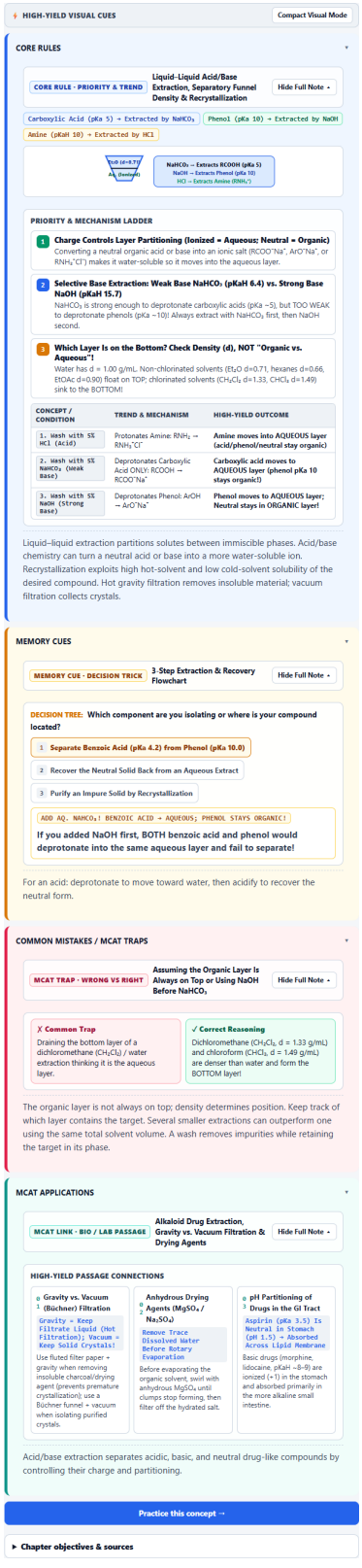
Screenshot: layout-mobile-light



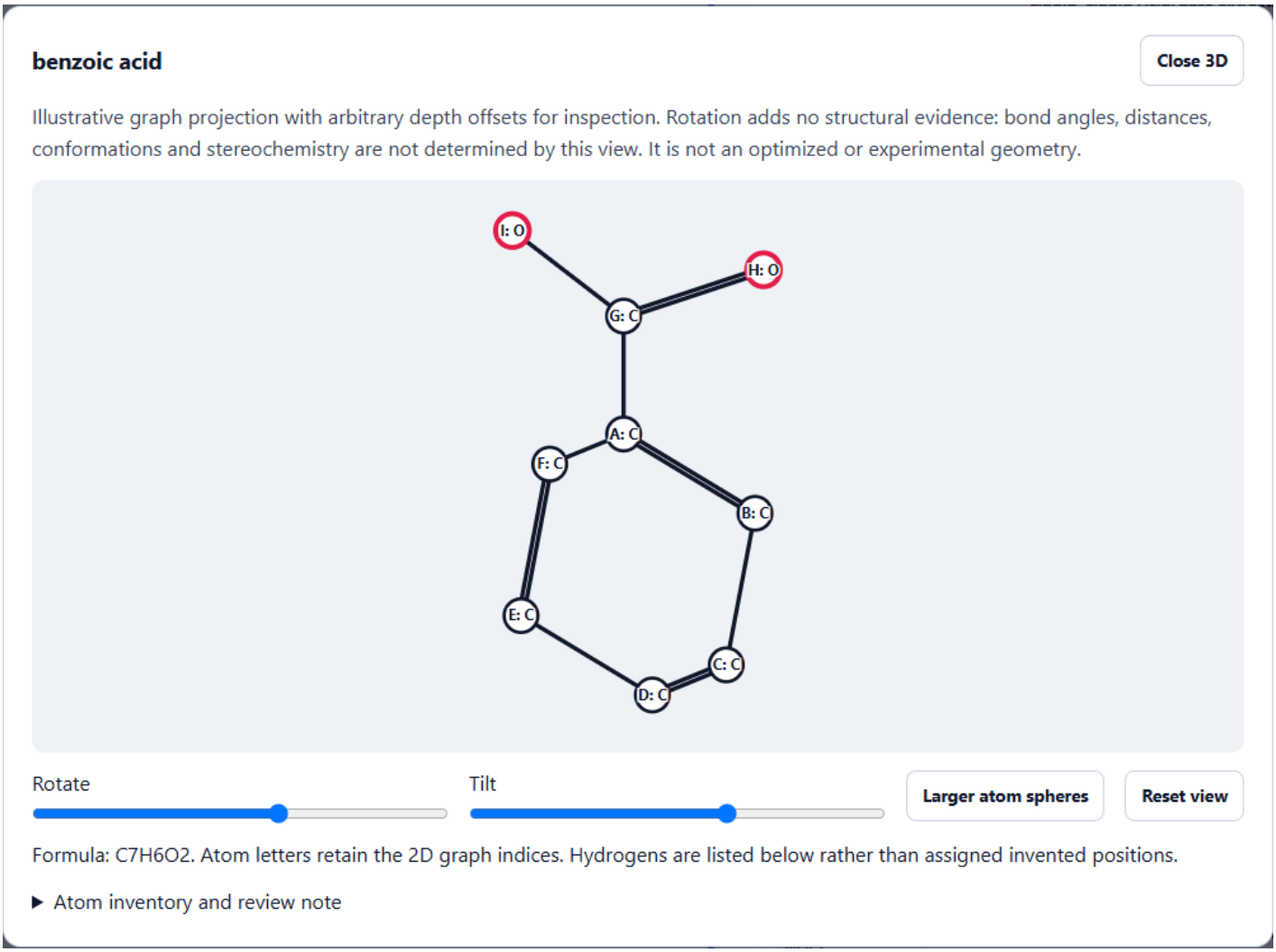
Screenshot: layout-mobile-dark



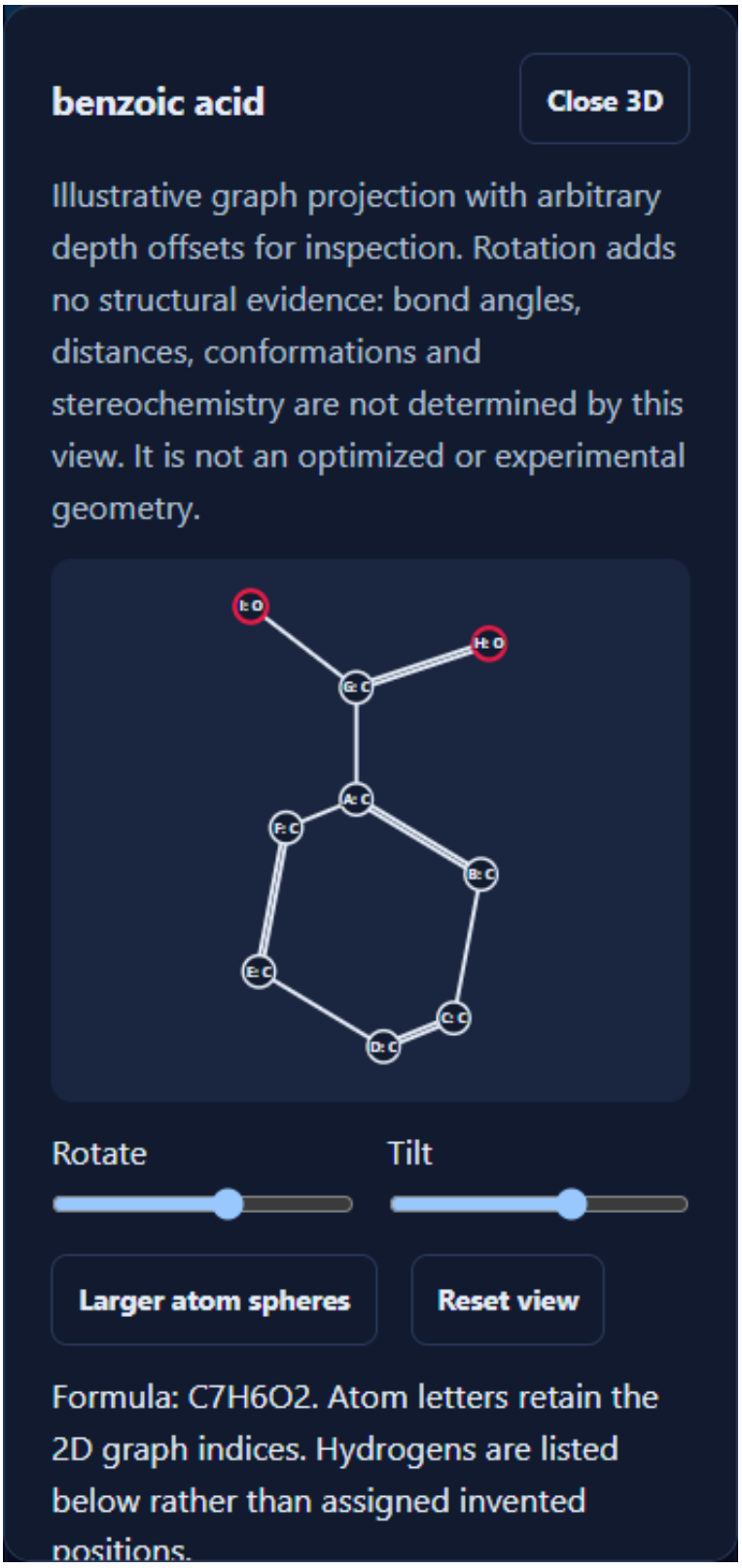
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

60 screenshots and 12 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.